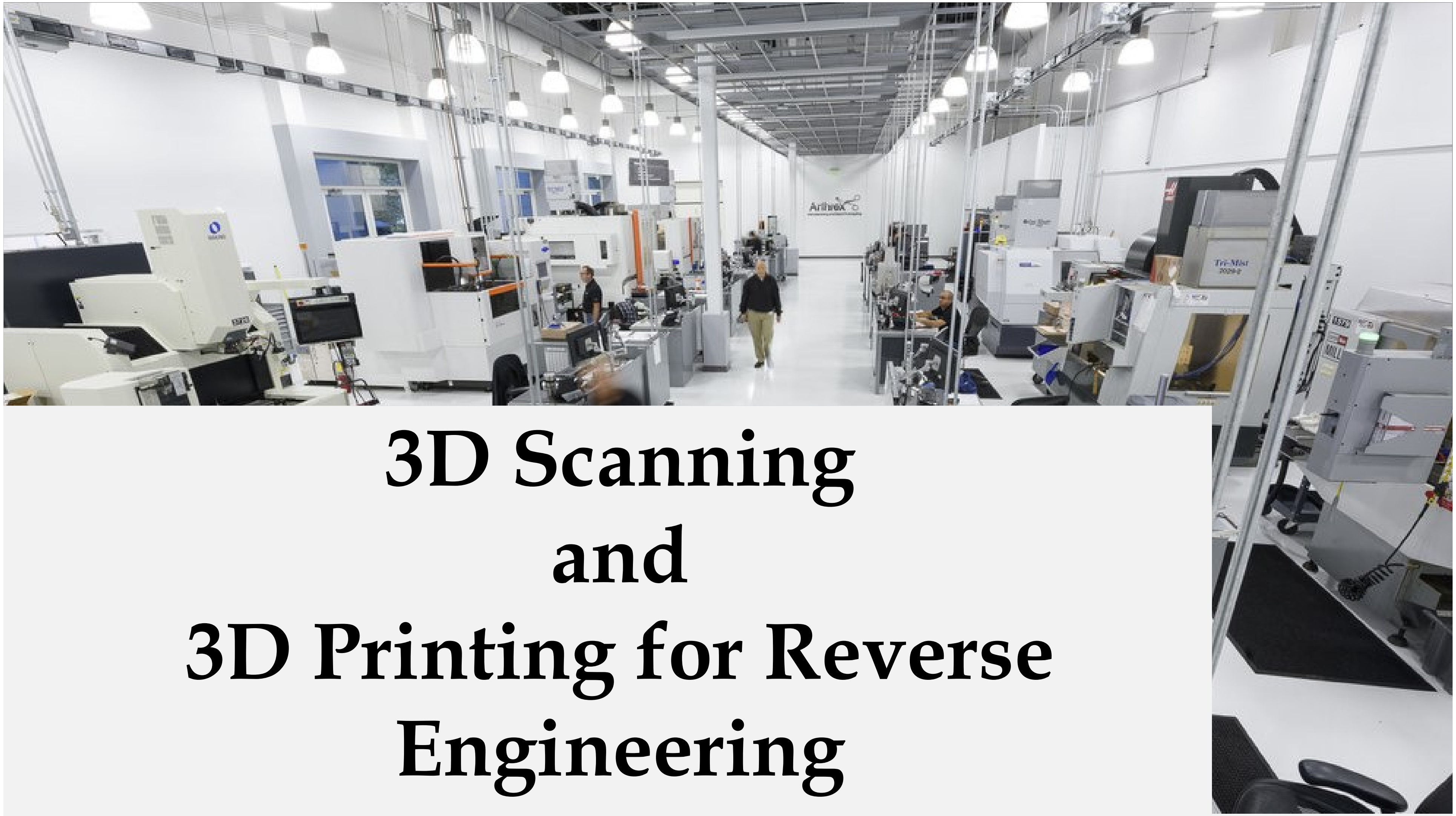
5ME42A(ME) – Manufacturing Technology

5th Semester B.E. Mechanical Engineering

Dr. Kailash Chaudhary
Ph.D. (Mechanical Design), M.E. (P & I), B.E. (Mechanical Engg)
Assistant Professor
Department of Mechanical Engineering
MBM University Jodhpur



3D Scanning
and
3D Printing for Reverse Engineering

OUTLINE

- Rapid Prototyping Process and Machines
- Applications
- Advantages and Limitations
- Case studies

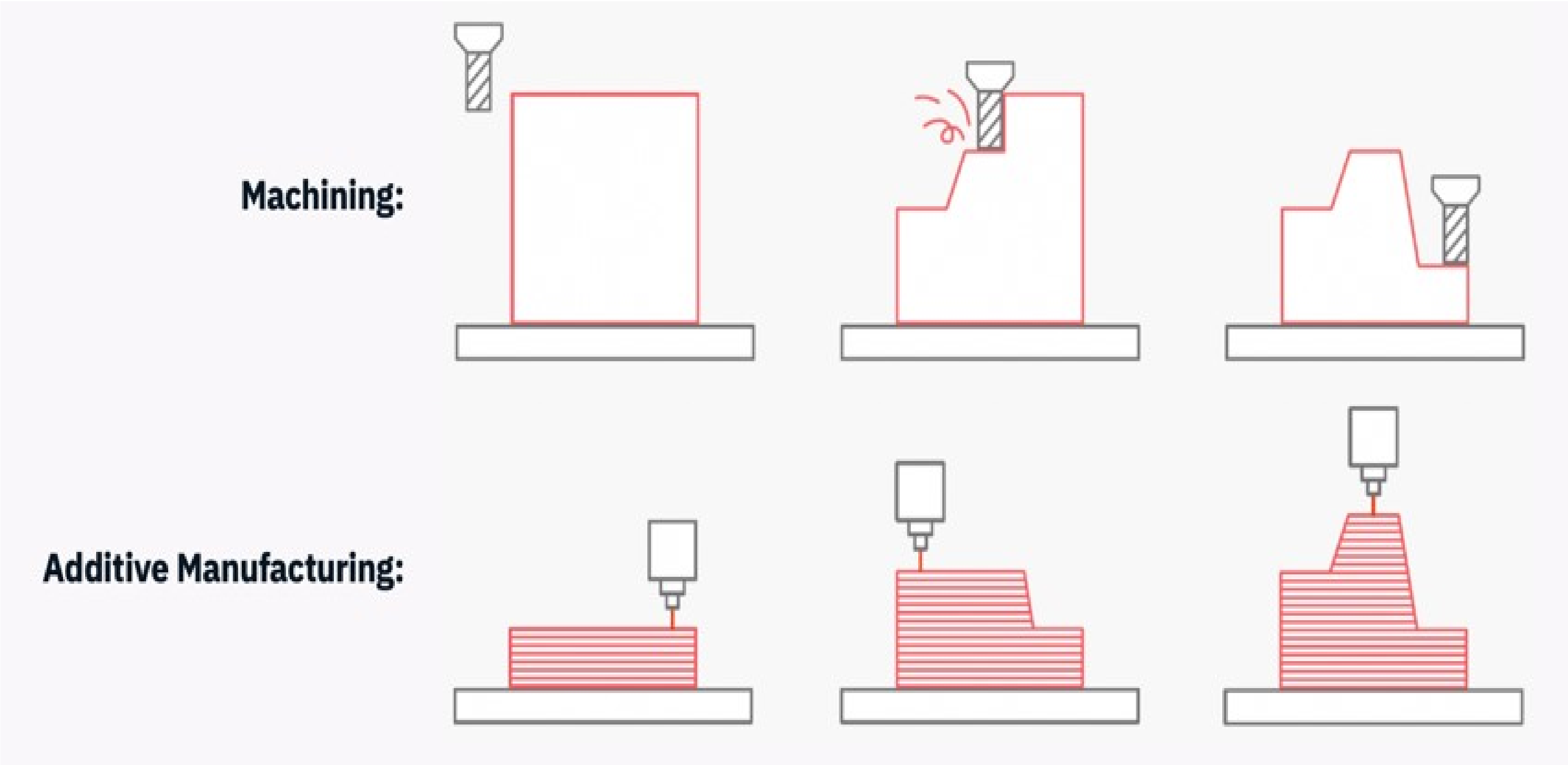
PROTOTYPING

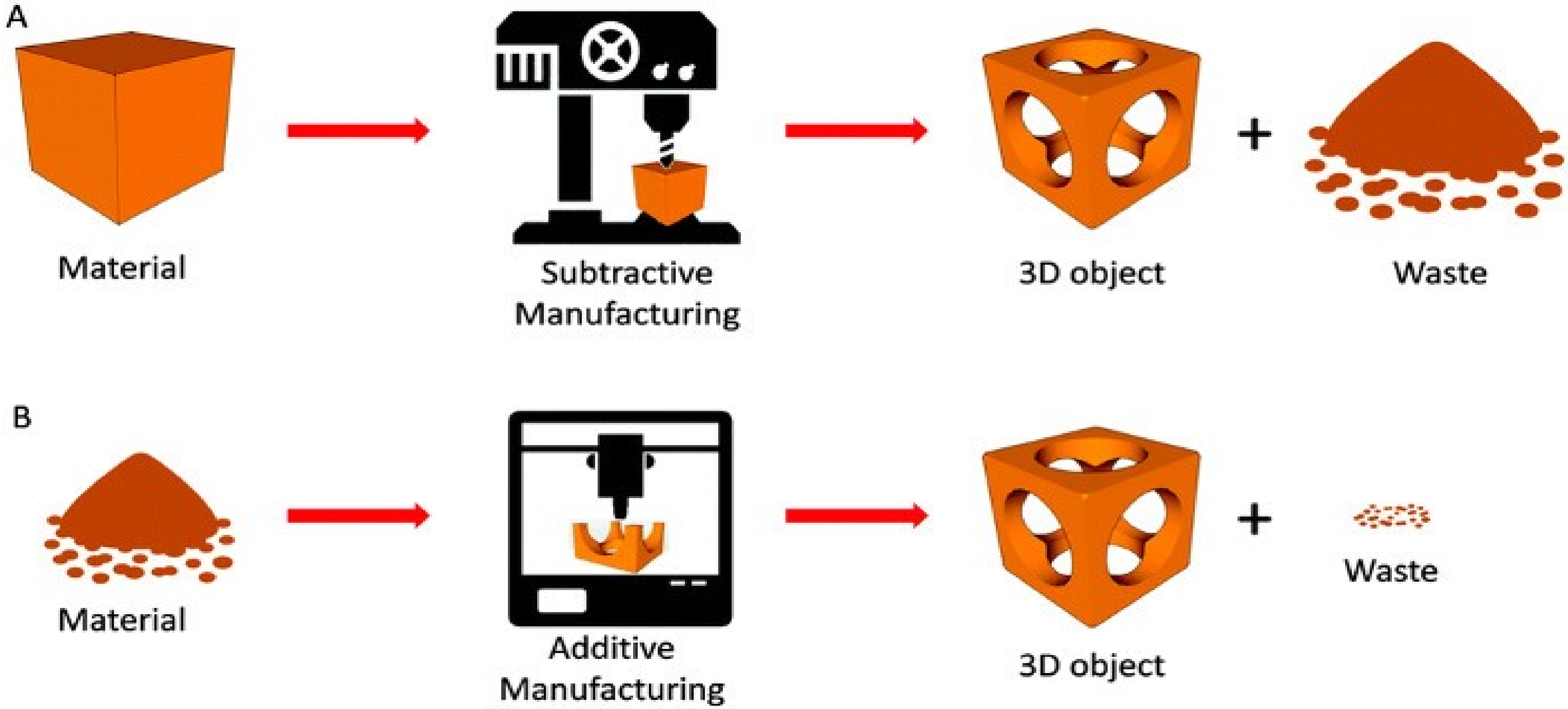
Prototype is a draft version or an approximation of final product.

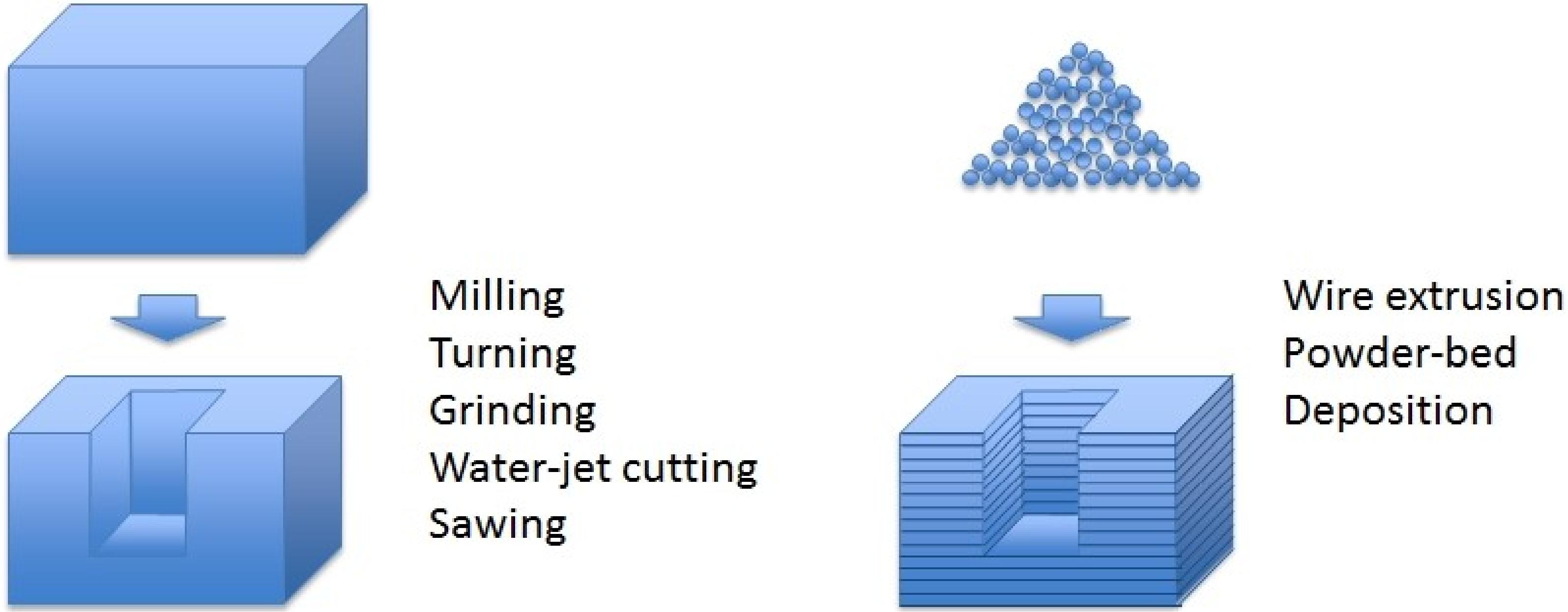
Essential part of product development process for assessing the performance and functionality of a design before a significant investment is made

RAPID PROTOTYPING

Technology which produces physical model in a layer by layer manner directly from their CAD models without any tools, dies and fixtures and also with little human intervention

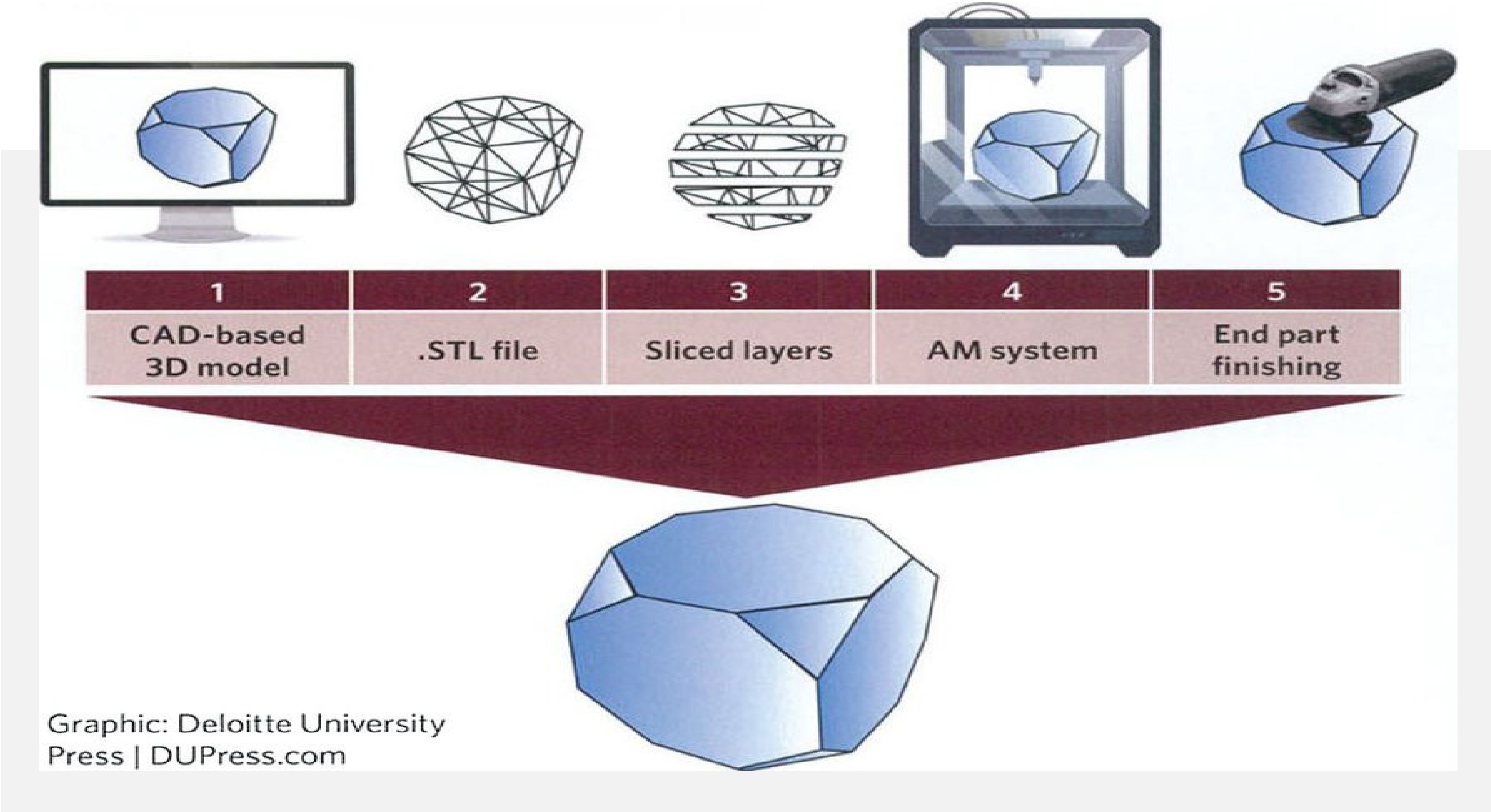

Machining:
Additive Manufacturing:

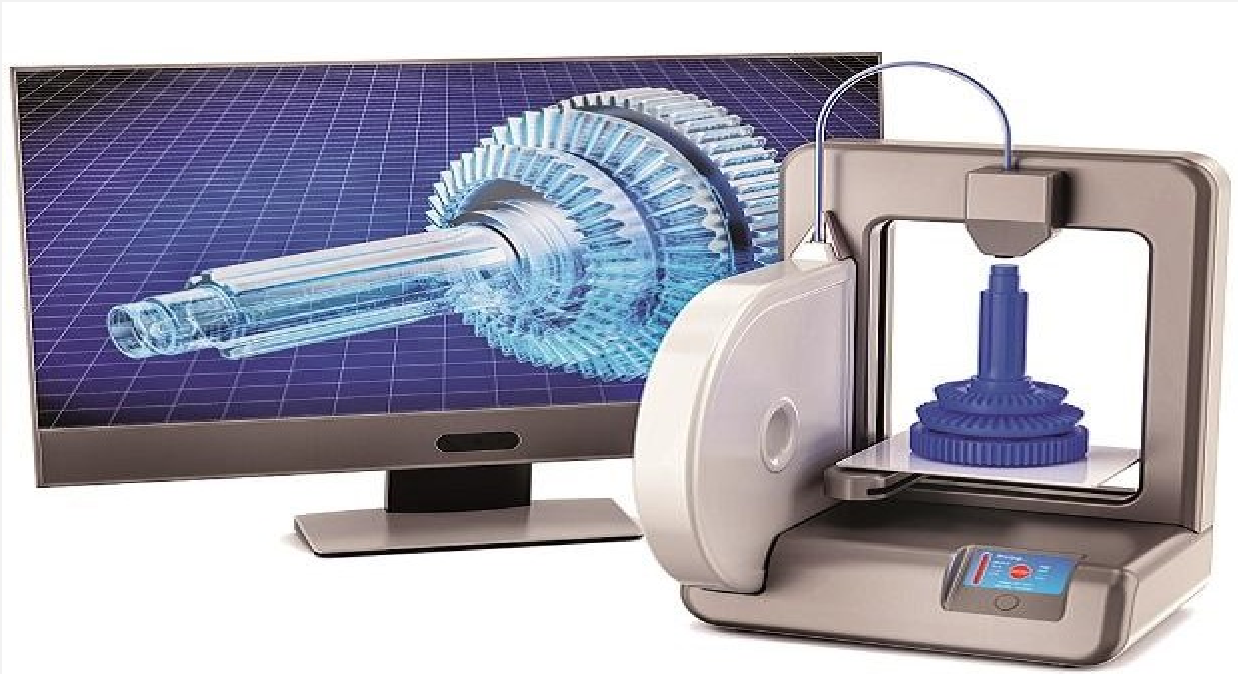

A
Material
Subtractive
Manufacturing
3D object
+
Waste
B
Material
Additive
Manufacturing
3D object
+
Waste



Subtractive machining removes material by physical or chemical processing

Additive manufacturing adds layers of material


1
CAD-based 3D model
2
.STL file
3
Sliced layers
4
AM system
5
End part finishing
Graphic: Deloitte University Press | DUPress.com



RAPID PROTOTYPING PROCESS

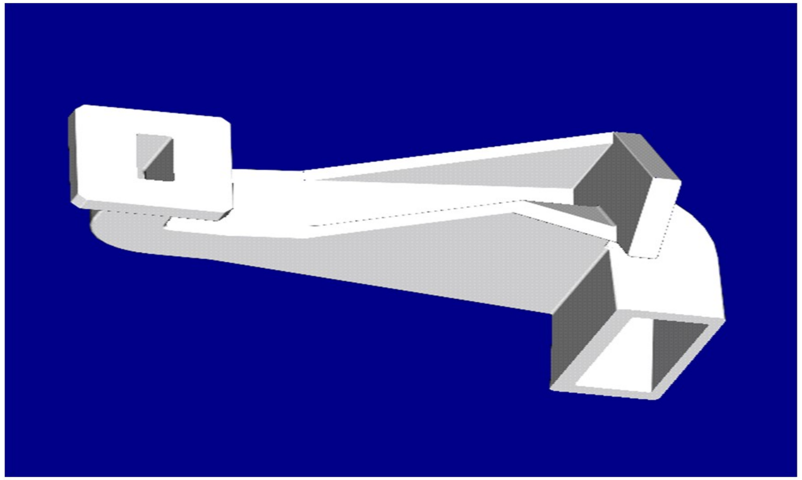
basic five-step of process are:

1. Create a CAD model of the design.
2. Convert the CAD model to STL format.
3. Slice the STL file into thin cross-sectional layers.
4. Construct the model one layer atop another.
5. Clean and finish the model.

Steps discussed in next few slides ➡

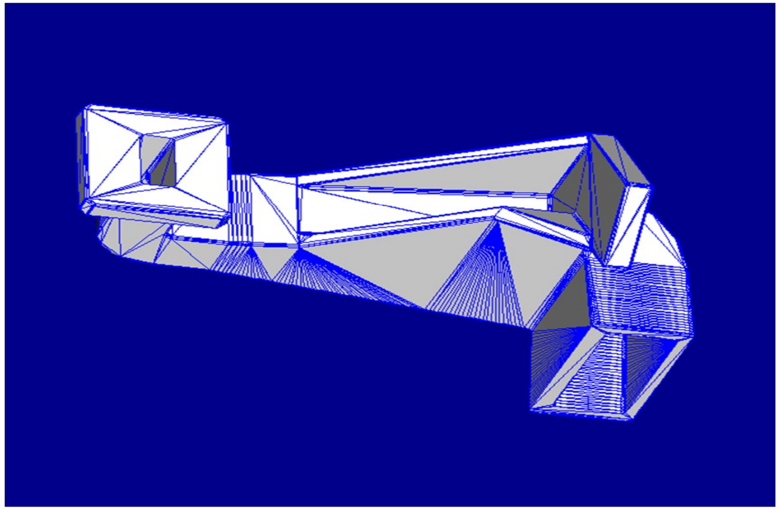
CAD Model Creation

- object to be built is modeled using a CAD software
- Solid modelers, such as Pro/ENGINEER



Conversion to STL Format

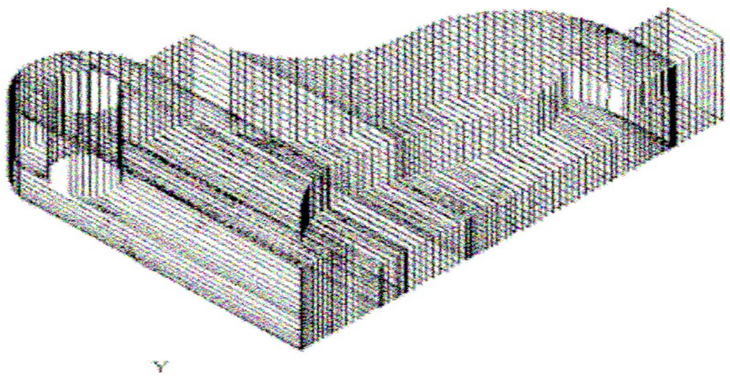
- STL format has been adopted as the standard of the rapid prototyping industry
- STL files use planar elements, they cannot represent curved surfaces exactly
- Increasing the number of triangles improves the approximation



STL is abbreviation of "stereolithography”

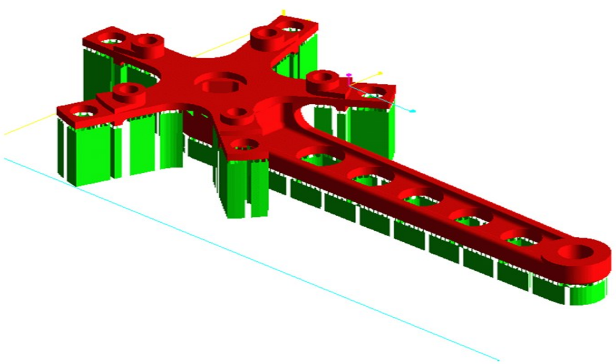
Slice the STL File

- a pre-processing program prepares the STL file to be built
- also generate an auxiliary structure to support the model during the build



Layer by Layer Construction

- build one layer at a time from polymers, paper, or powdered metal
- layers of thickness from 0.01 mm to 0.7 mm thick are created



TYPES OF RAPID PROTOTYPING SYSTEM

Based on the form of the starting material:

- **LIQUID-BASED RAPID PROTOTYPING SYSTEMS**
- **SOLID -BASED RAPID PROTOTYPING SYSTEMS**
- **POWDER -BASED RAPID PROTOTYPING SYSTEMS**

LIQUID-BASED RAPID PROTOTYPING SYSTEMS

video

- ✓ Starting material is a liquid
- ✓ Mostly resins and polymers.

Includes the following processes:

- ❑ Stereo lithography
- ❑ Solid ground curing
- ❑ Droplet deposition manufacturing

SOLID-BASED RAPID PROTOTYPING SYSTEMS

video LOM
video FDM

- ✓ Starting material is a solid wood, plastic, metal sheets etc.

Solid-based RP systems include the following processes:

- ❑ Laminated object manufacturing
- ❑ Fused deposition modeling

POWDER-BASED RAPID PROTOTYPING SYSTEMS

video

- ✓ Starting material is a powder of hard materials like Stainless steel
- ✓ Non-ferrous metals such as Bronze; Elastomers; Composites

Powder-based RP systems include the following:

- ❑ Selective laser sintering
- ❑ Three dimensional printing
- ❑ Laser engineered and Net shaping

COMMON TYPES OF RAPID PROTOTYPING SYSTEMS:

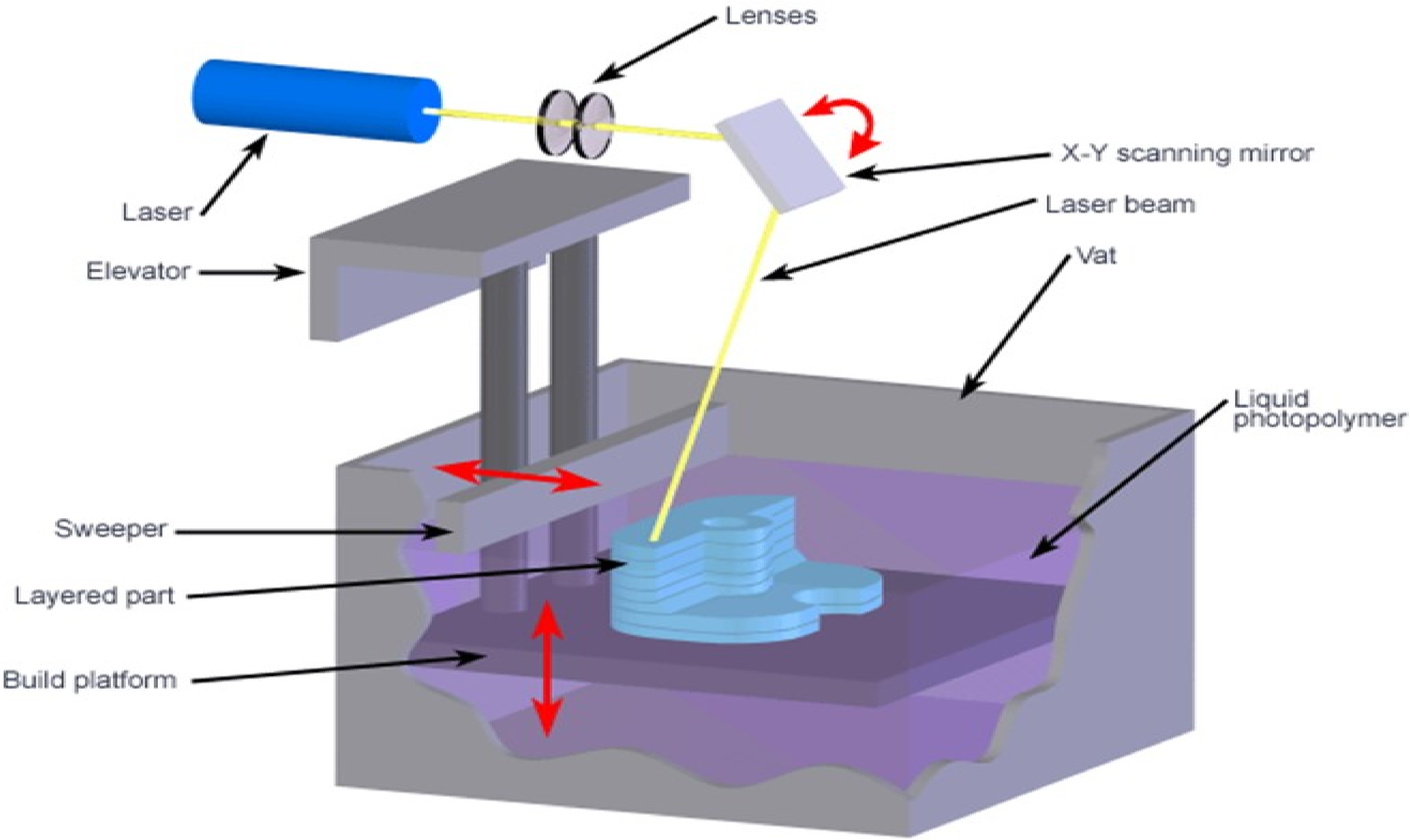
- **SLA (Stereo Lithography)**
- **SLS (Selective Laser Sintering)**
- **LOM (Laminate Object Manufacturing)**
- **FDM (Fused Deposition Modeling).**

Different technologies use different materials to produce the parts

Steps discussed in next few slides

SLA (Stereo Lithography)

1. Builds 3D model from liquid photo sensitive polymers when exposed to UV rays.
2. A low power highly focused UV laser is used for solidification
3. An elevator incrementally lowers the platform into the liquid polymer.
4. Process is repeated until prototype is complete.
5. Model is placed in an UV oven for complete curing.

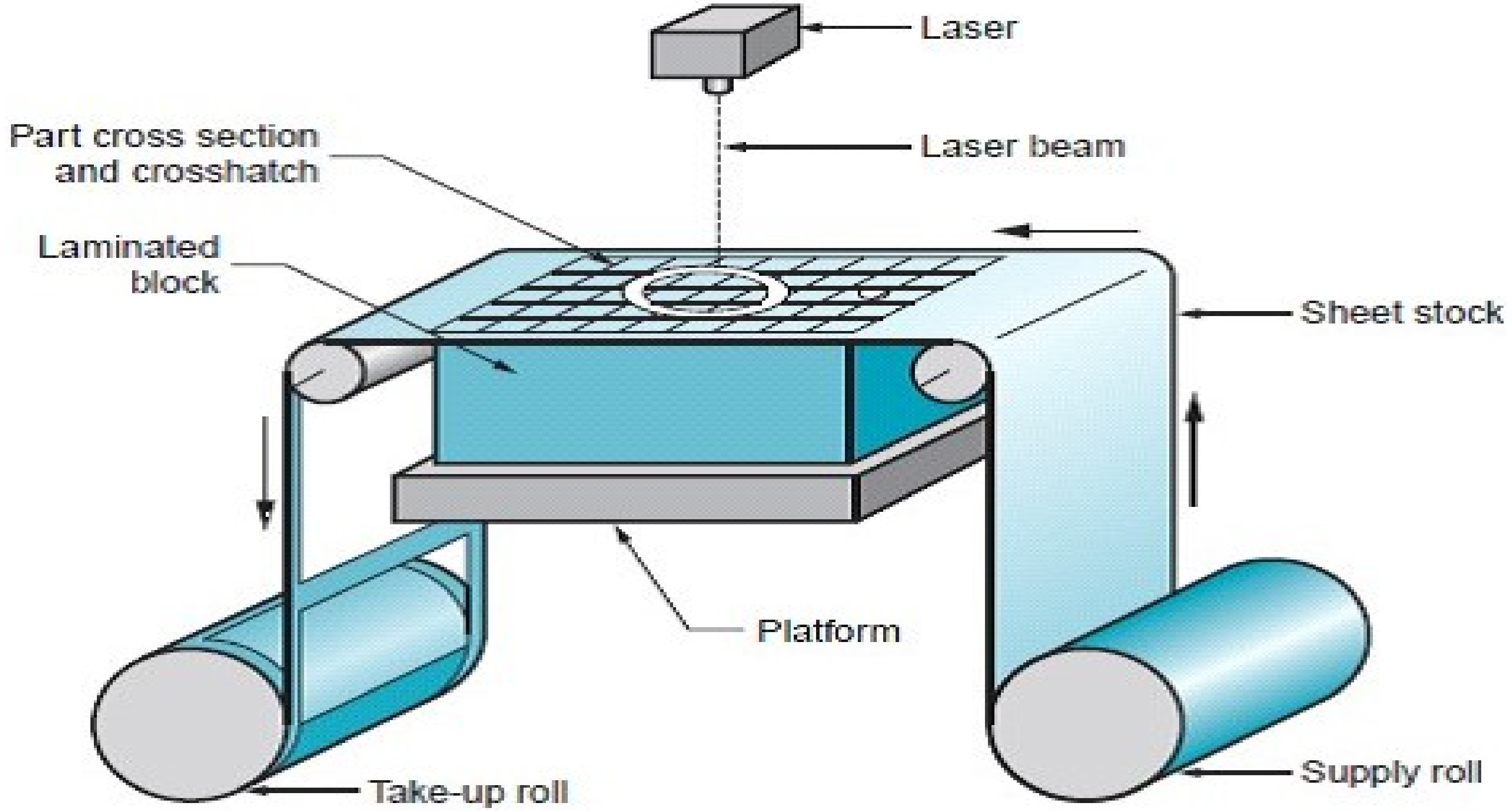


Lenses
X-Y scanning mirror
Laser beam
Laser
Elevator
Vat
Liquid
photopolymer
Sweeper
Layered part
Build platform

LOM (Laminated Object Manufacturing)

1. Thin slices of material (usually paper or wood) are successively glued together to form a 3D shape.
2. Process uses two rollers to control the supply of paper with heat-activated glue to a building platform.
3. When new paper is in position, it is flattened and added to the previously created layers using a heated roller.
4. The shape of the new layer is traced and cut by a blade or a laser. When the layer is complete, the building platform descends and new paper is supplied.

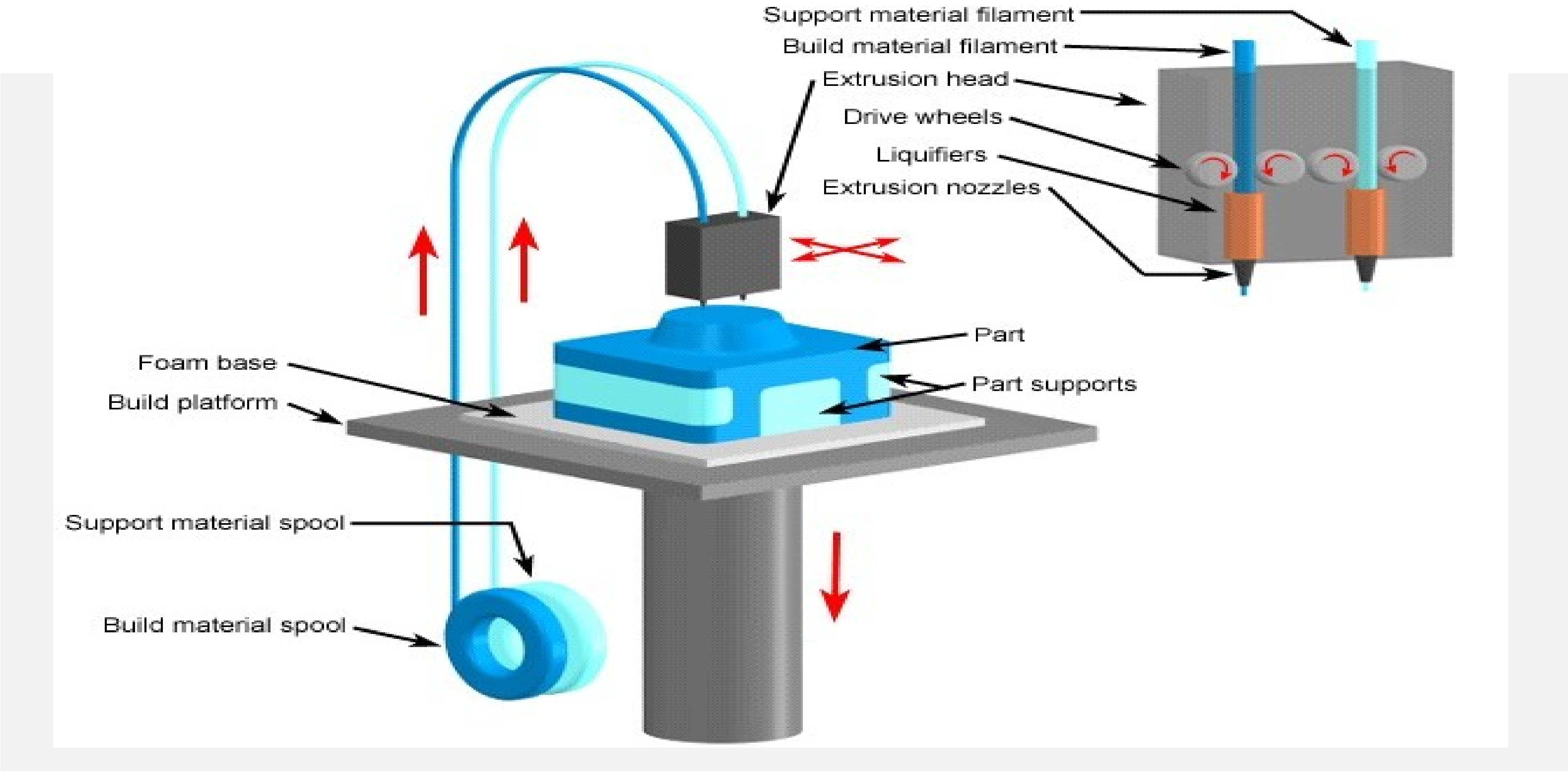
LOM (Laminated Object Manufacturing)



FDM (Fused Deposition Modelling)

1. Extruder head moves in two principle directions over a table
2. Table can be raised or lowered as needed
3. Thermo plastic or wax filament is extruded through the small orifice of heated die
4. Extruder head follows a predetermined path from the file
5. After first layer the table is lowered and subsequent layers are formed .

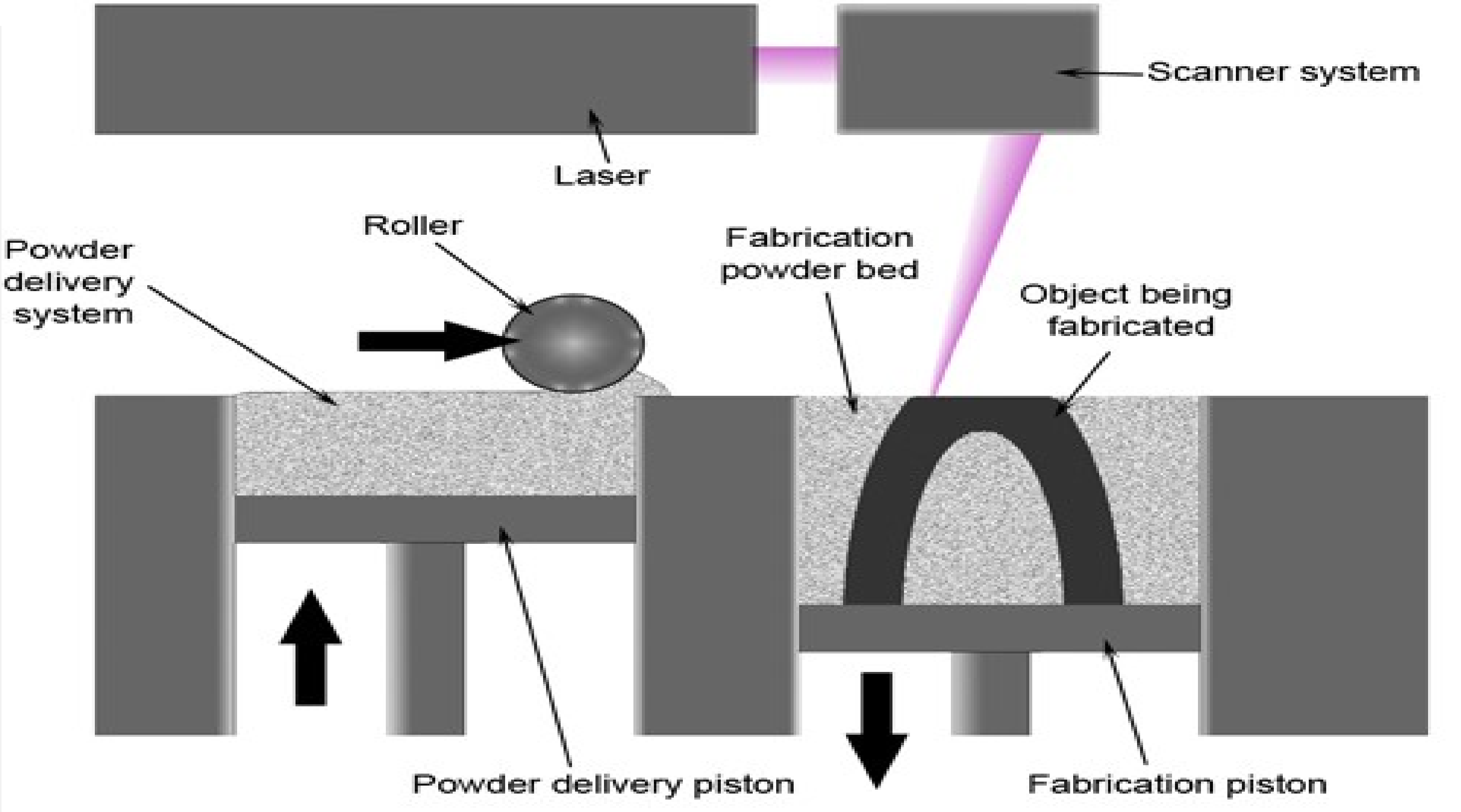
FDM (Fused Deposition Modelling)



SLS (Selective Laser Sintering)

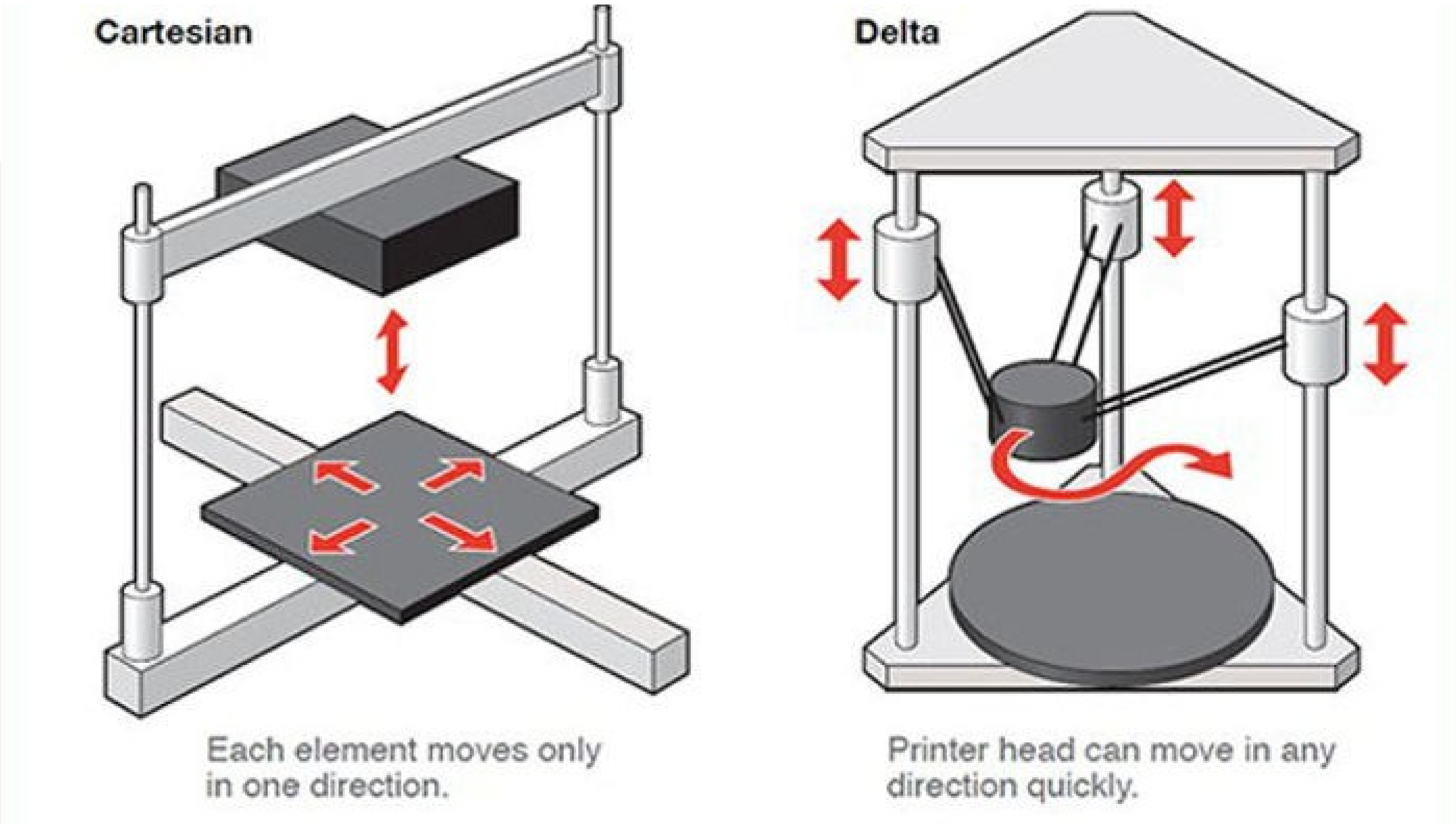
1. **Powdered material is supplied to the building area by a roller.**
2. **For each layer, a laser traces the corresponding shape of the part on the surface of the building area, by heating the powder until it melts, fusing it with the layer below it.**
3. **The roller moves the new material to the building platform, leveling the surface, and the process repeats.**

SLS (Selective Laser Sintering)



GENERAL CLASIFICATION OF FDM

1. CARTESIAN FDM 3D PRINTING
2. POLAR FDM 3D PRINTING
3. DELTA FDM 3D PRINTING
4. FDM 3D PRINTING WITH ROBOTIC ARMS


Cartesian
Each element moves only in one direction.
Delta
Printer head can move in any direction quickly.

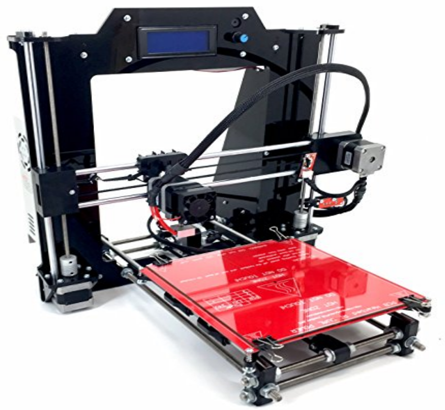
DIFFERENT TYPES OF FDM AVAILABLE IN MARKET

Flashforge 3D Printer Creator Pro

Lulzbot Mini Desktop 3D Printer

Maker Gear M2 Desktop 3D Printer

LulzBot TAZ 6 3D Printers



DIY RepRap Guru Prusa I3

Dremel Idea Builder 3D Printer

MATERIALS USED IN FDM

1. POLYLACTIC ACID (PLA)
2. ACRYLONITRILE BUTADIENE STYRENE (ABS)
3. POLYAMIDE (PA)
4. HIGH IMPACT POLYSTYRENE (HIPS)
5. THERMOPLASTIC ELASTOMERS (TPE)
6. OTHERS

Advantages of Rapid Prototyping

- ❖ Process is fast and accurate.
- ❖ Superior quality surface finish is obtained.
- ❖ Separate material can be used for component and support
- ❖ No need to design jigs and fixtures.
- ❖ No need of mould or other tools.
- ❖ Post processing include only finishing and cleaning.
- ❖ Harder materials can be easily used .
- ❖ Minimum material wastage.
- ❖ Reduces product development time considerably

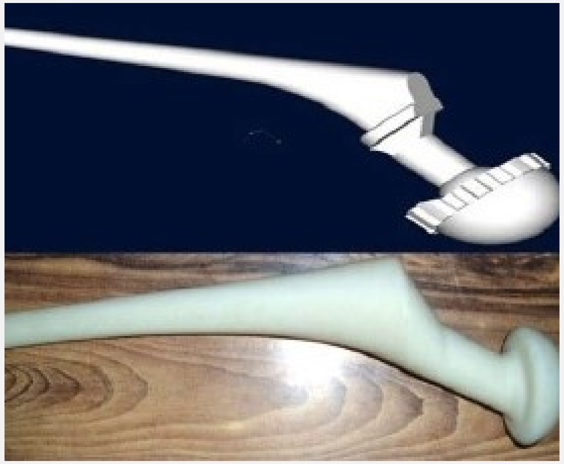
Applications of Rapid Prototyping

RAPID TOOLING

- Patterns for Sand Casting
- Patterns for Investment Casting
- Pattern for Injection moldings

RAPID MANUFACTURING

- Short productions runs
- Custom made parts
- Manufacturing of very complex shapes



BIOMEDICAL APPLICATIONS

- Customized surgical implants
- Mechanical bone replicas
- Forensics

Automobile – Interior Components

- A dashboard insert for a radio or the control elements of the air conditioning is displayed. laser sintering, polyamide

- Fuel tank, laser sintering, polyamide

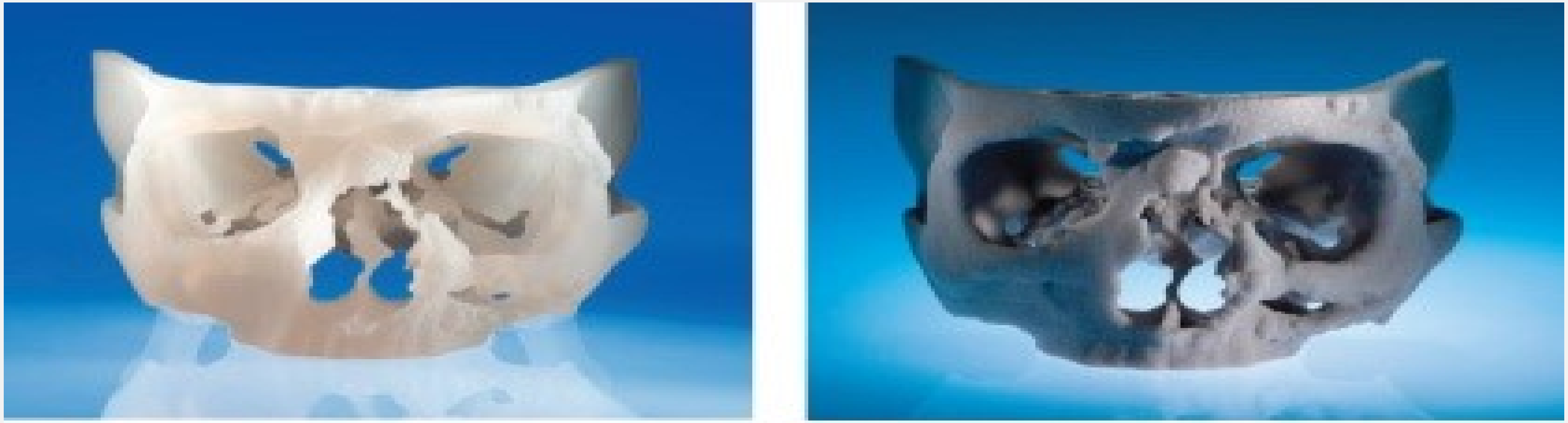
Automobile – Exterior Components

- Modified front spoiler for a vehicle's special edition

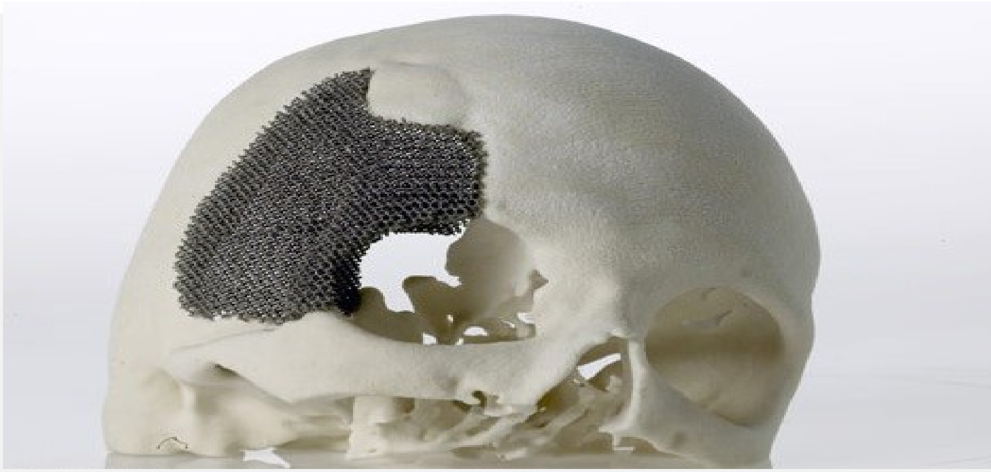
The first 3D printed car “Strati”

The 3D printed bus “Olli”

Applications of AM in Medical Engineering



- 3D printed facsimiles of a skull based on identical CT data: 3D printing (left); stereolithography (right)



- Skull models with individual implants: facsimile of a skull, manufactured by stereolithography (left) and by laser sintering (right).

- Hip socket was manufactures by SLM, titanium

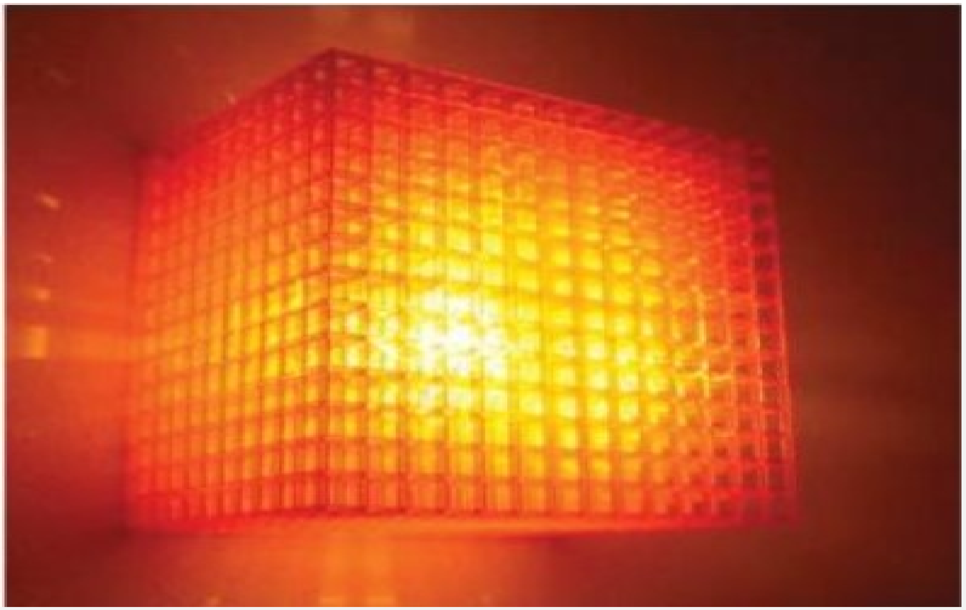
Applications of AM in Aerospace Engineering

- Hot-air duct for an aviation turbine: 3D printing (metal)

- 3D printed lightweight bracket, for the Airbus A350 XWB; SLM, titanium

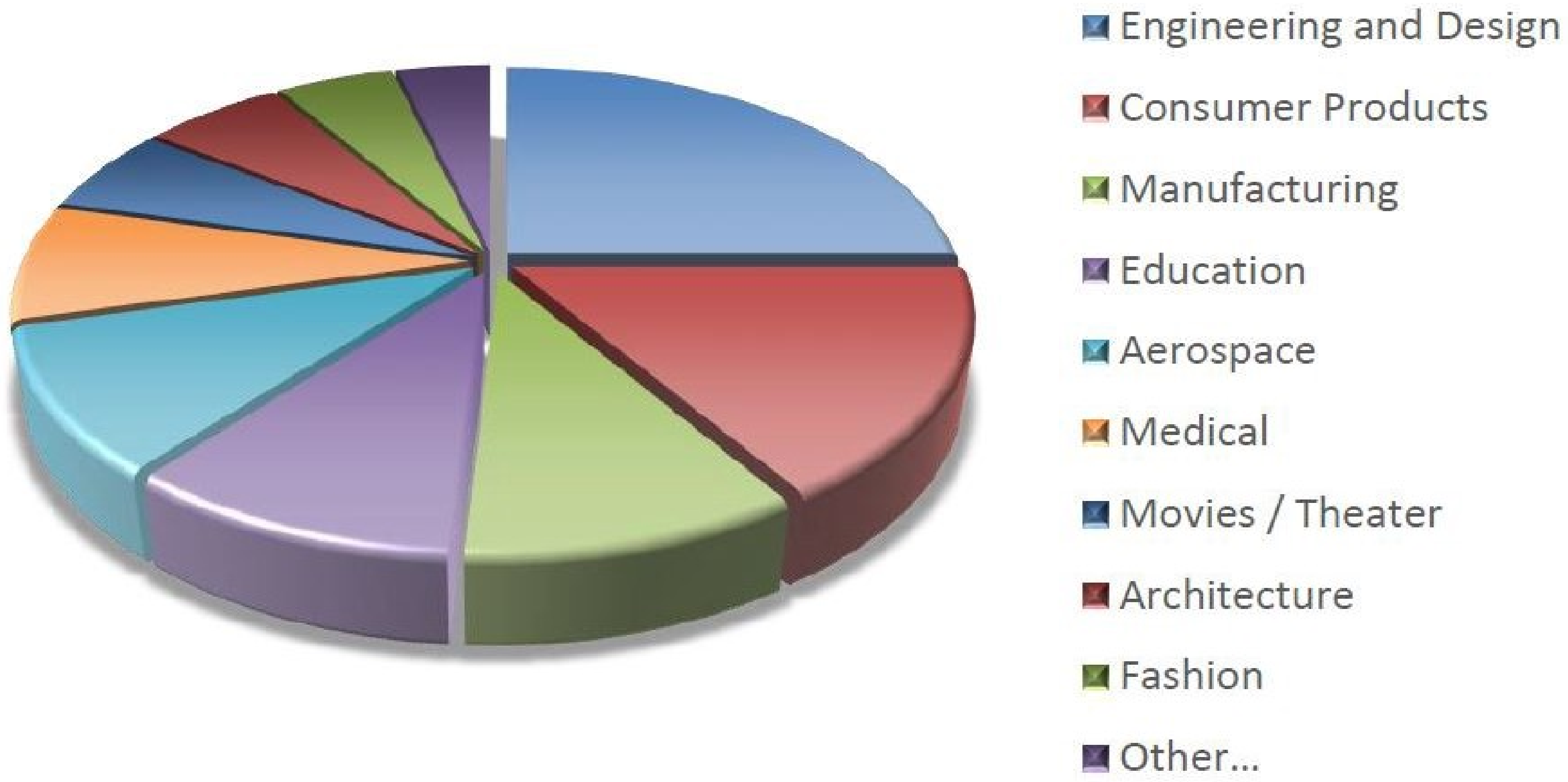
Applications of AM in Consumer Goods

- Cocktail cup: laser stereolithography



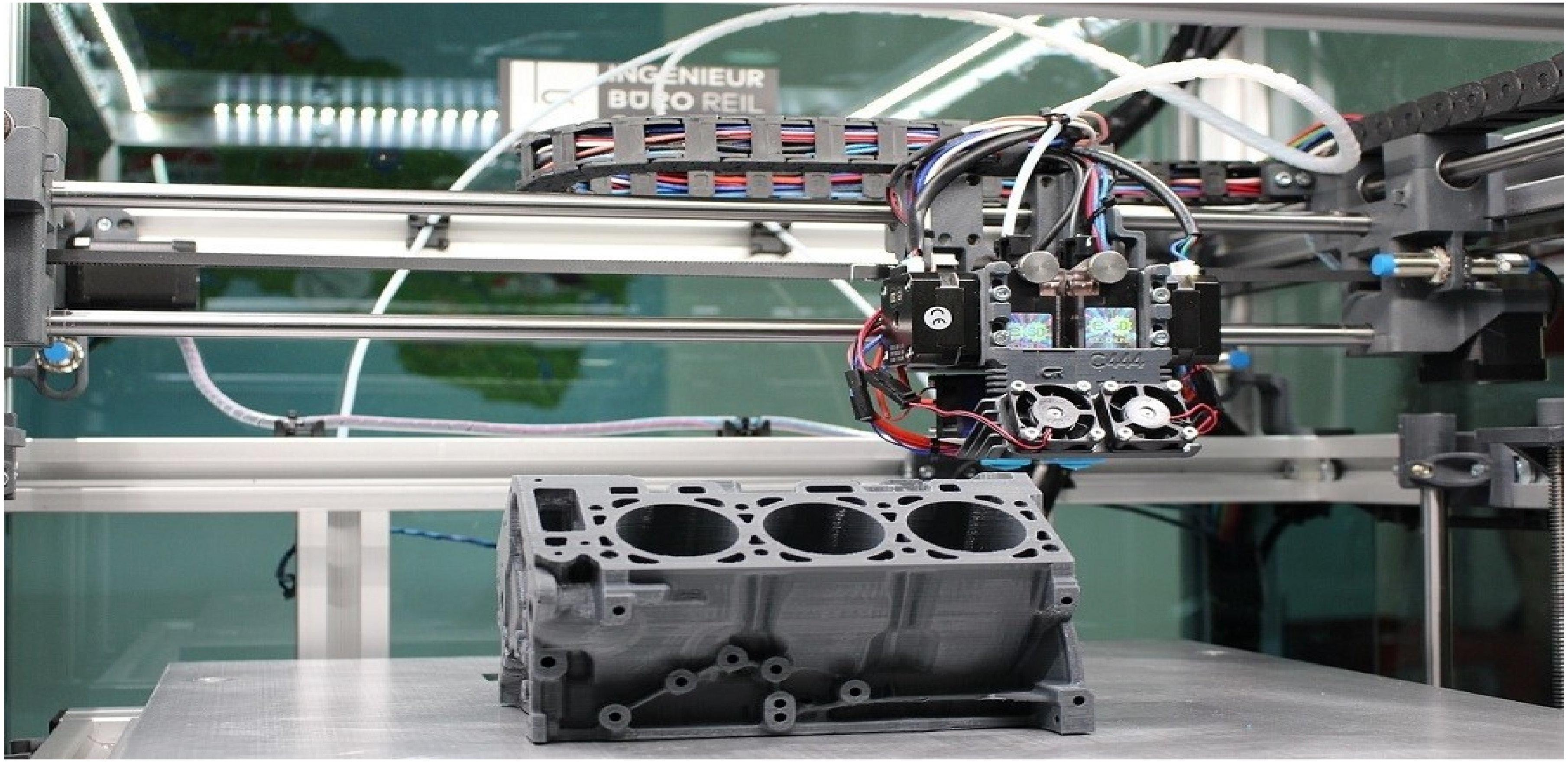
- Lamp: laser sintering, polyamide (left); lamp: laser stereolithography, epoxy resin (right)

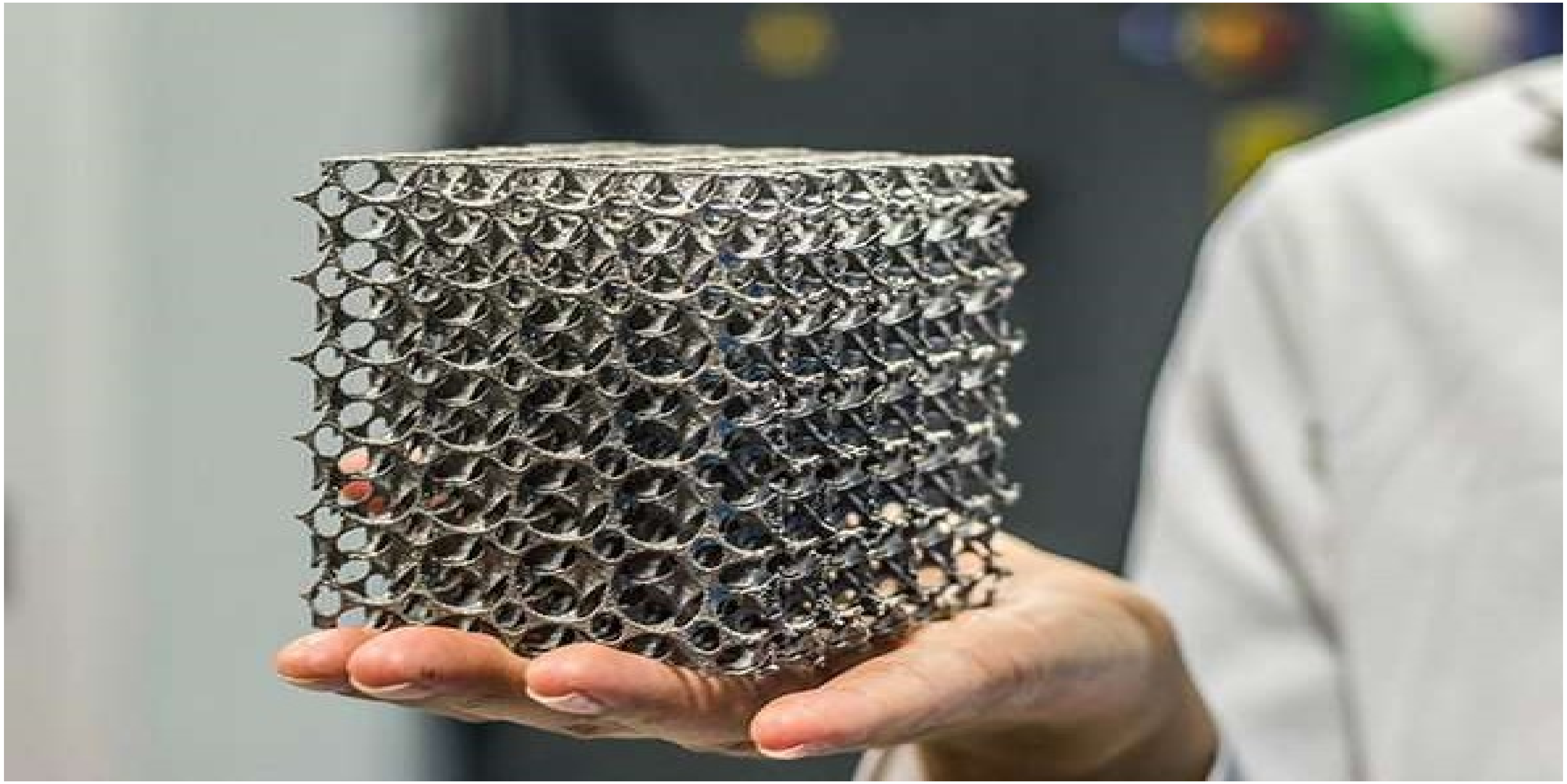

3D printer

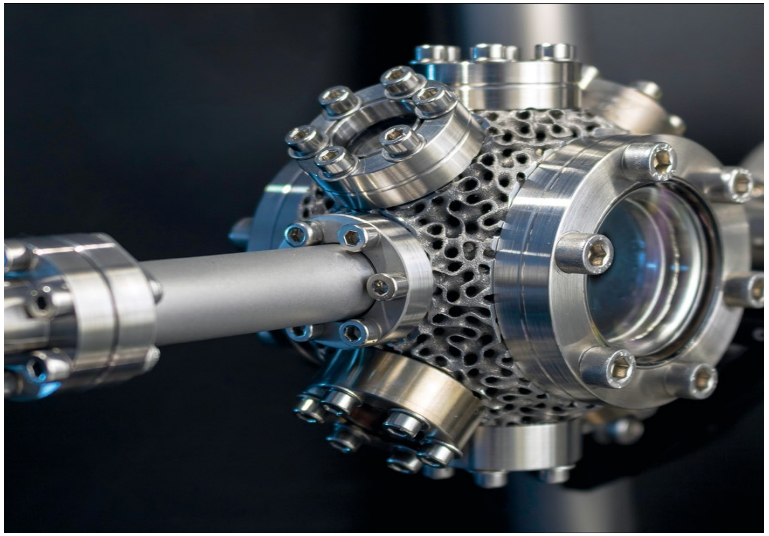

Engineering and Design
Consumer Products
Manufacturing
Education
Aerospace
Medical
Movies / Theater
Architecture
Fashion
Other...




3D printing use in fighting COVID 19


INGENIEUR
BÜRO REIL






5 cm

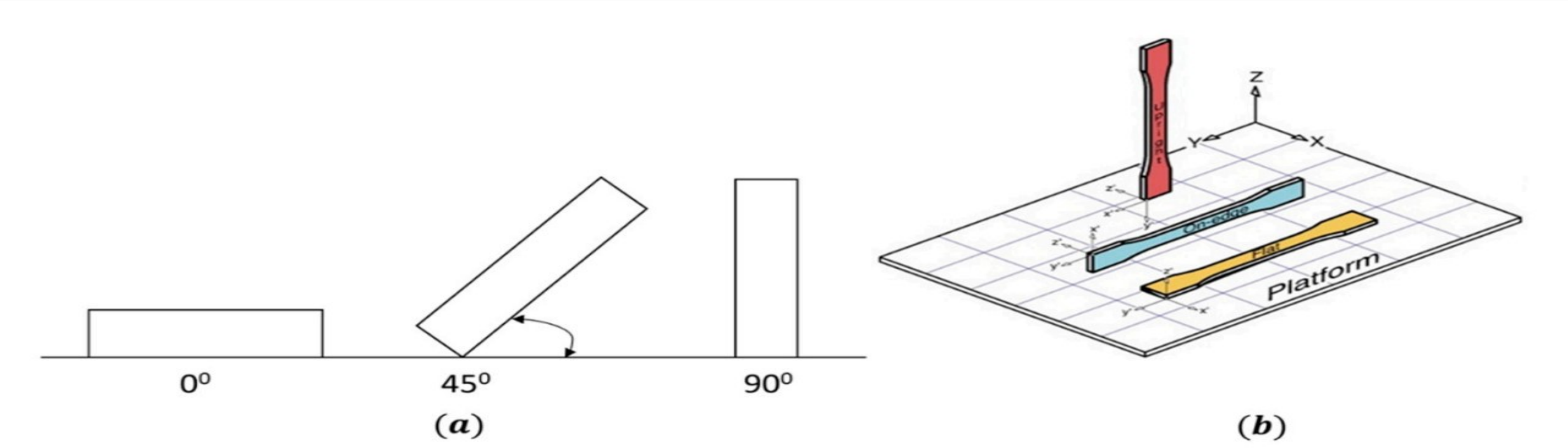

SOLS
i.materialise
SECURED 3D
3D printer security
NEW MATTER
MakerBot INDUSTRIES
dreamforge
cgtrader
MakieLab
THINGIVERSE
formlabs
AUTHENTISE
shapeways
GRABCAD
Makible
makexyz
sculpteo
stilnest

Process Parameters of FDM Process

- Build orientation
- Layer thickness
- Air gap
- Rater angle
- Raster width
- Infill pattern, Infill density

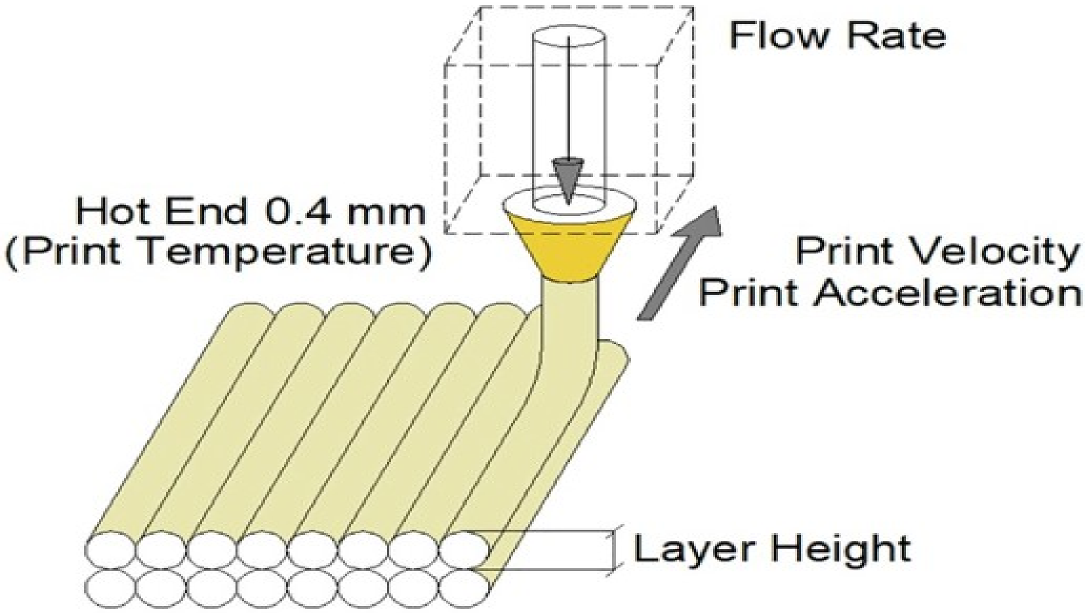
Build orientation

It is defined as orientation of the part on the build platform with respect to X-, Y- and Z- axis.



Layer thickness

- It is define as height of each deposited layer. This is calculated with respect to vertical axis i.e. Z- axis. It is mainly depends upon diameter of the extruder nozzle and generally less than the diameter of nozzle.

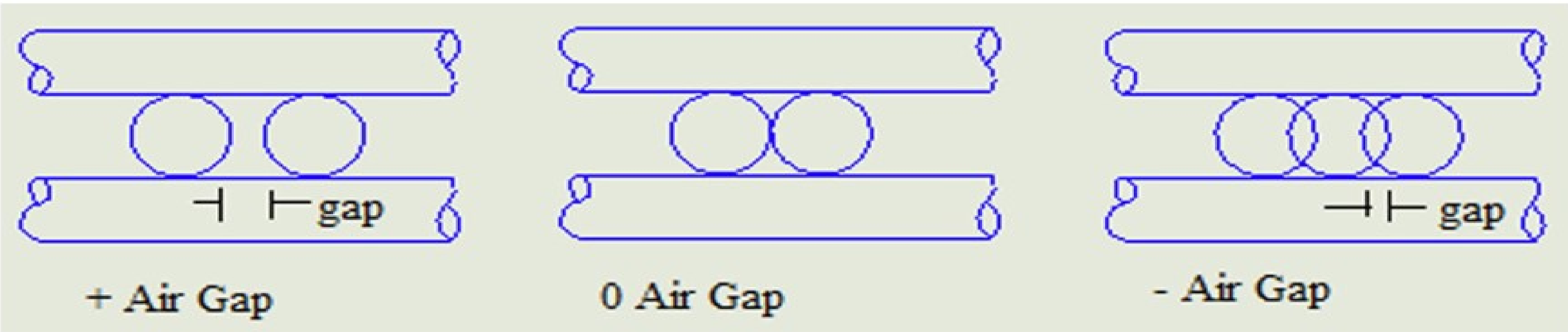


Extrusion temperature

- It is the temperature on which thermoplastic filament material is heated during fabrication process.
- It is mainly depends upon the material type and printing speed of process.
- Extrusion temperature affects the parts characteristics.

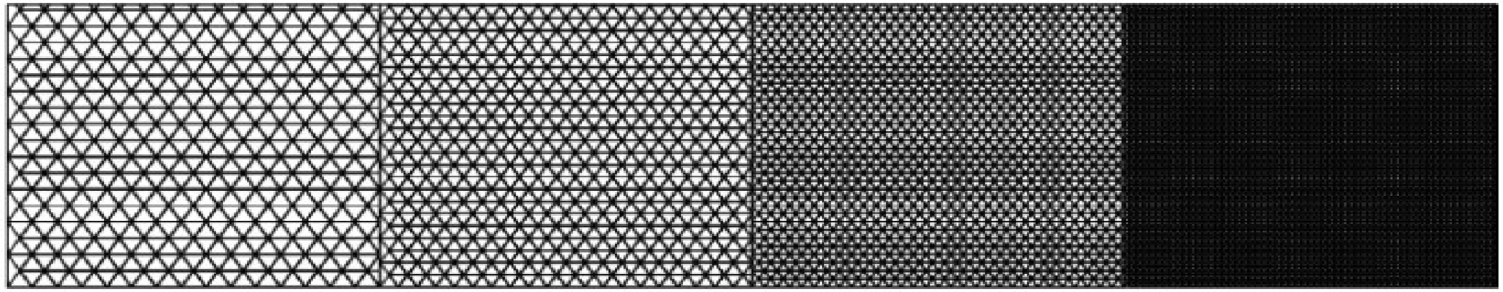
Air gap

- Defined as gap between two adjacent deposited layers.
- Air gap value can be positive, zero or negative.

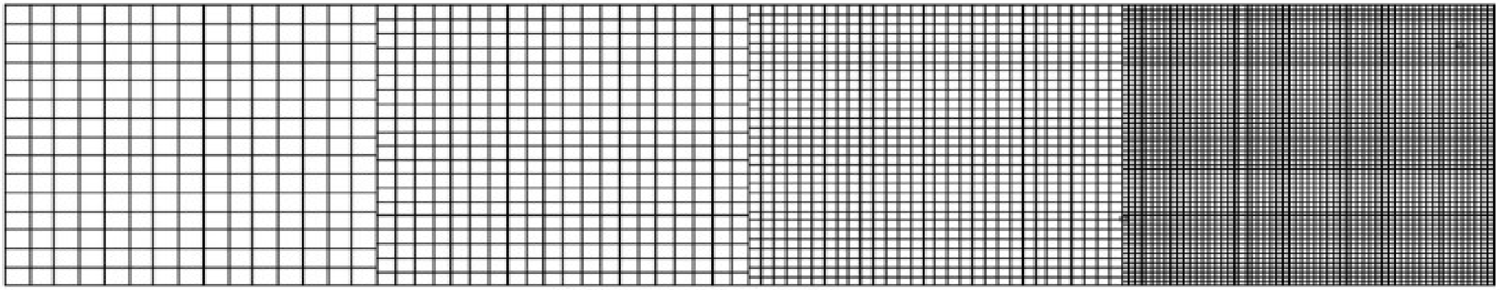


Infill density and Infill pattern

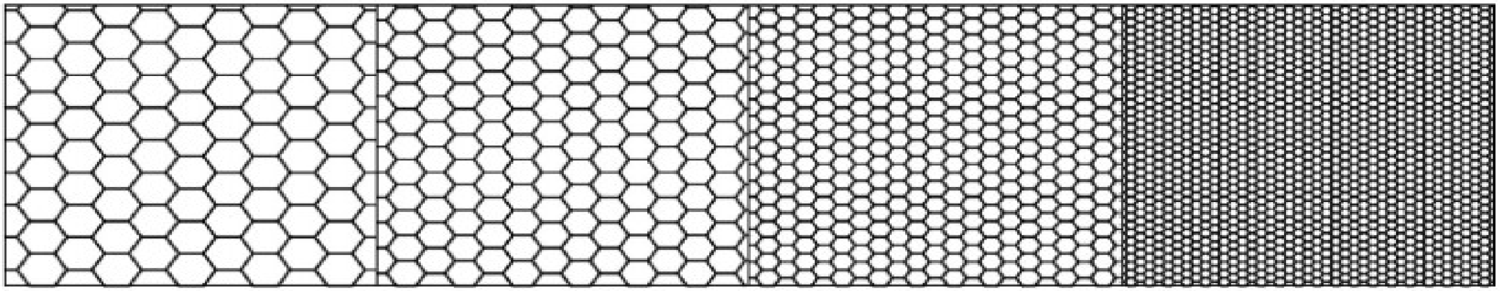
- Infill density is define as the internal structure density of 3-D printed object.
- The outer layer is in the solid form but inner part which is covered by outer layer, which has different shape and size and patterns are known as infill.
- The strength and mass of FDM fabricated product is depend upon infill density.
- Hexagonal, linear and diamond shape type infill patterns are generally adopted.



(a) Triangluar Infill



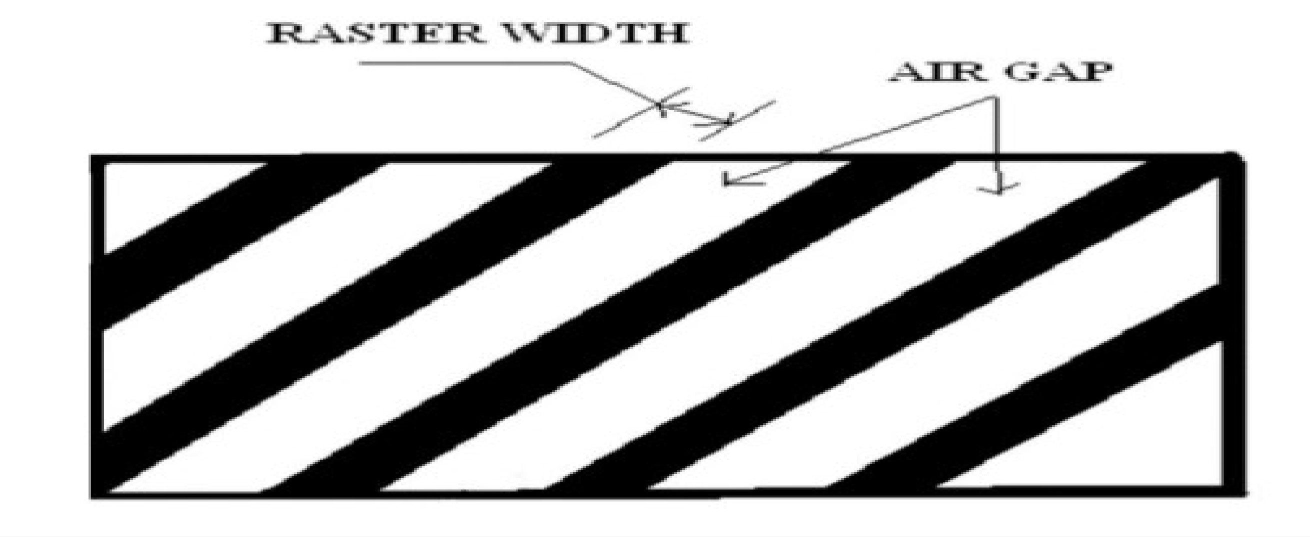
(b) Rectangluar Infill



(c) Honeycomb Infill

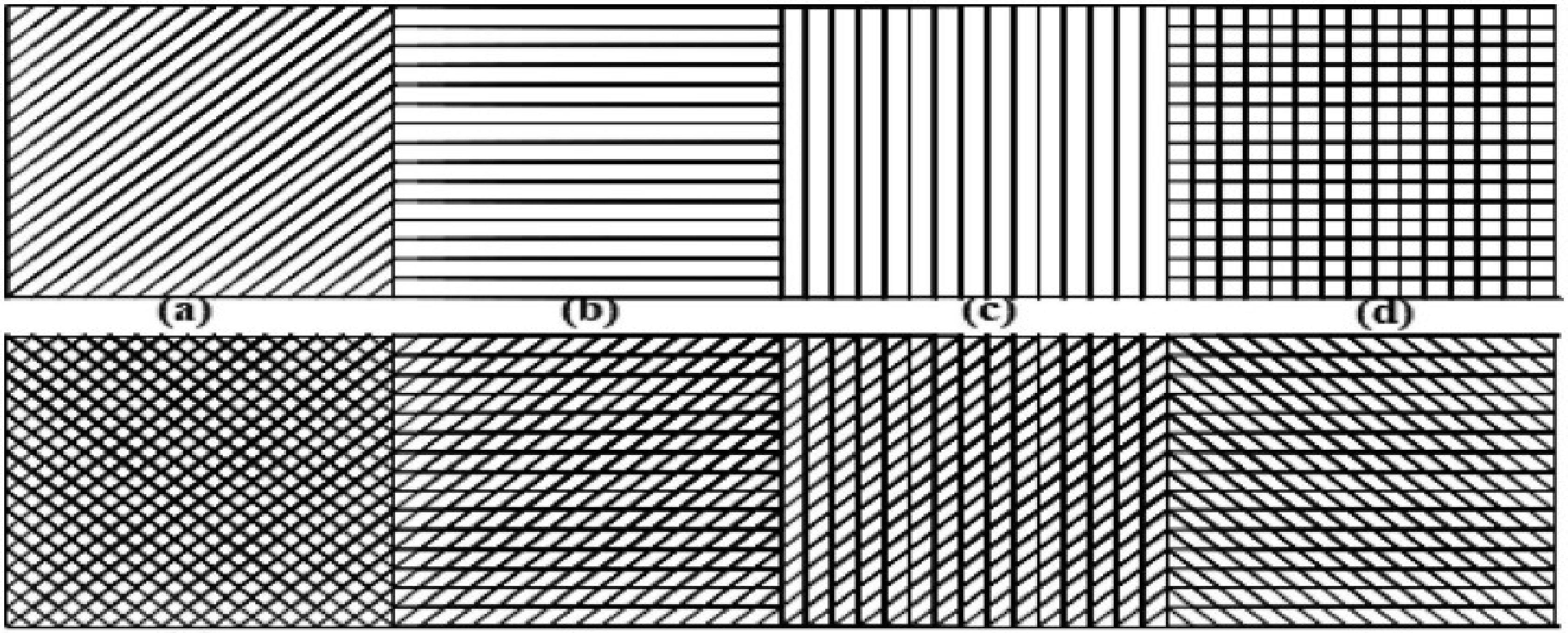
Raster Width

- It is defined as the width of the deposition beads.
- It depends on the extrusion nozzle diameter.



Raster orientation

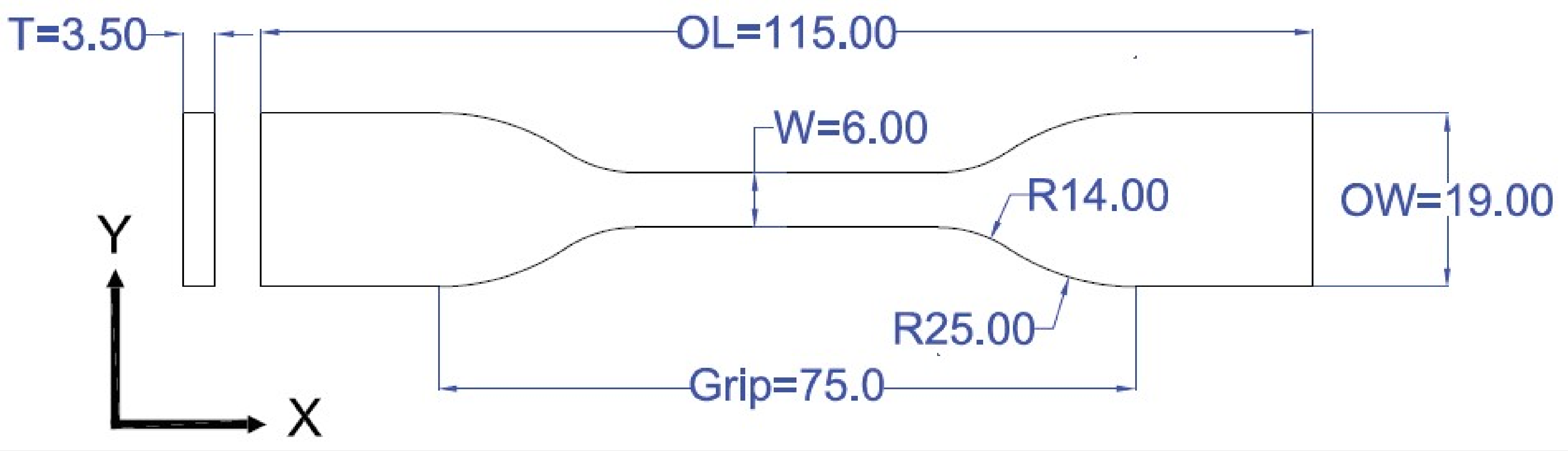
- Direction of the deposition beads with respect to the X – axis of the build platform of FDM machine is known as raster orientation.



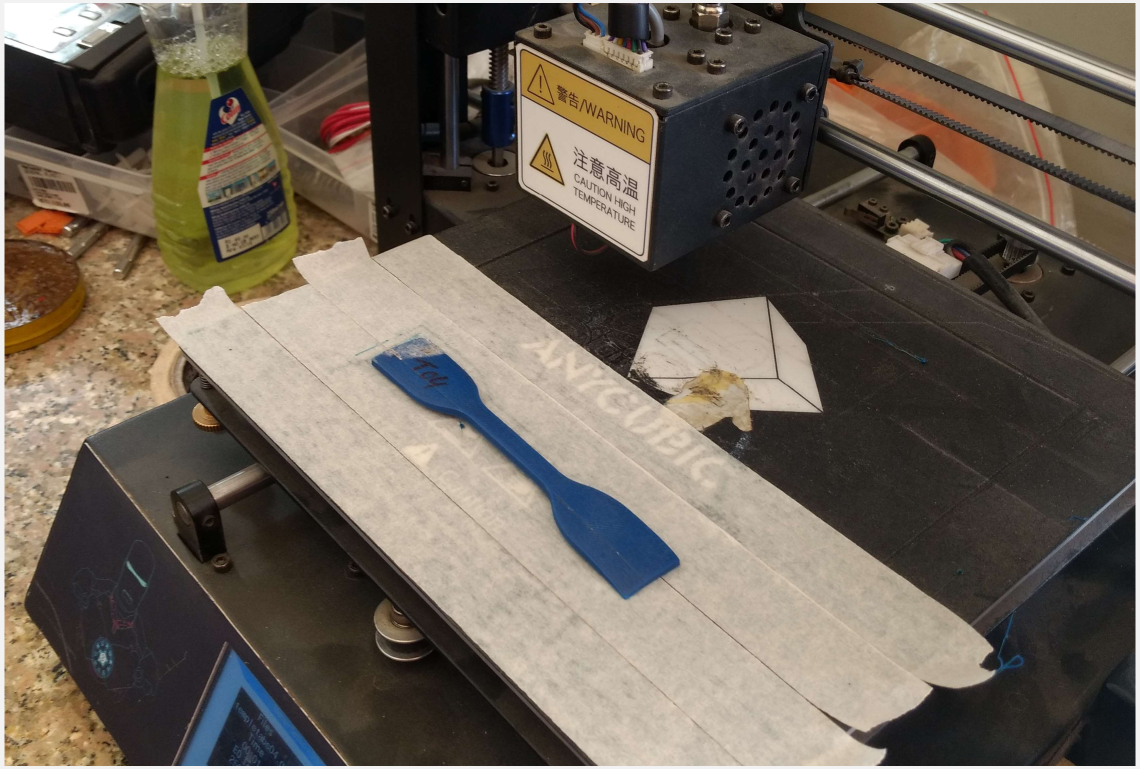
(a) (b) (c) (d)

(e) (f) (g) (h)

Created specimens' CAD model, dimensions in mm
ASTM 638 TYPE IV



SAMPLES PREPARED ON FDM MACHINE



CONSTANT PARAMETERS

Nozzle diameter 1.75mm

Extrusion Temperature

Abs 235 c

Heating Bed temperature 85 c

CONSTANT PARAMETERS

Nozzle diameter 1.75mm

Extrusion Temperature

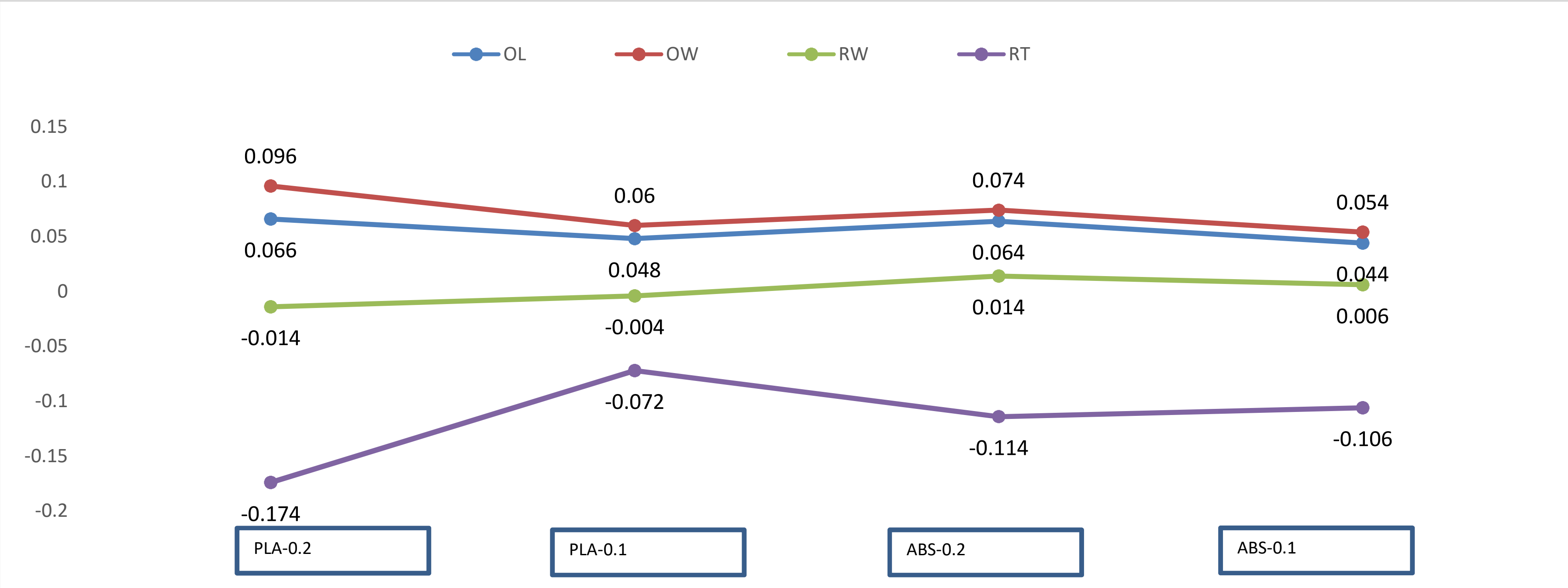
PLA 210 c

Heating Bed temperature 65 c

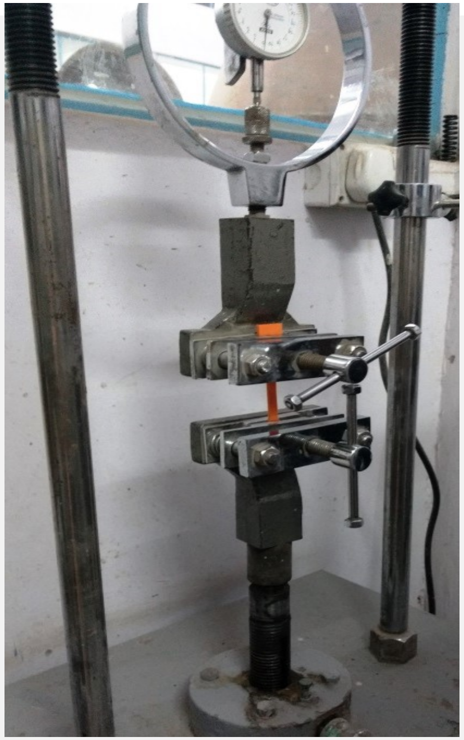
ABS AND PLA MATERIAL SAMPLES PREPARED USING FDM REPRAP MACHINE

DIMENSIONS ACCURACY TEST RESULTS

Graph between dimensional errors (OL, OW, RW, RT) and layer height (0.1, 0.2) for abs and pla material

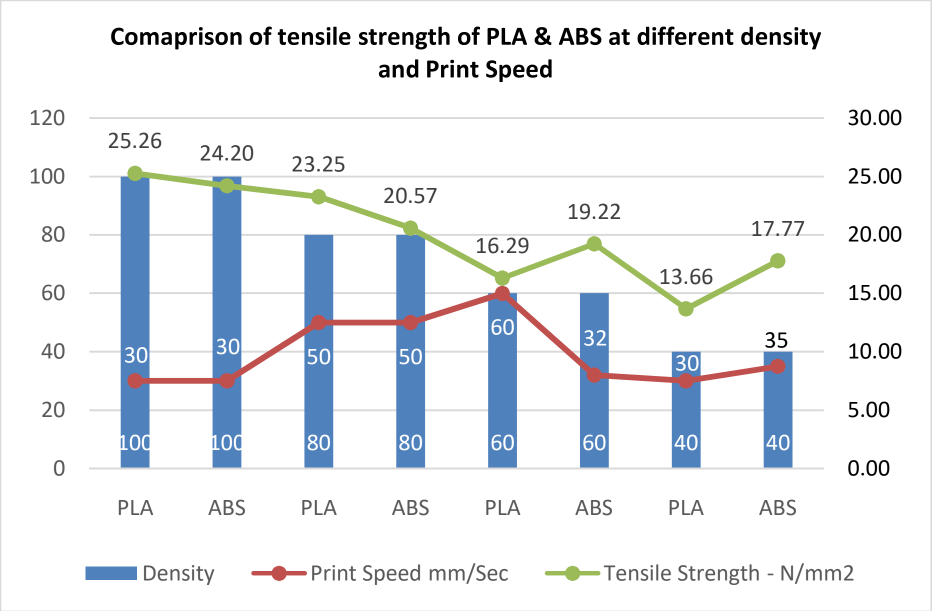


Tensile Testing Machine



Comparing the tensile strength of pla & abs material by varying density and print speed at layer height 0.2 mm

Material	Sample	Density %	Print Speed mm/sec	Extrusion Temperature ºC	Layer height mm	Cross Section Area mm²	Load in N	Tensile Strength - N/mm2
PLA	T-01	100	30	210	0.2	16.72	422.34	25.26
ABS	T-011	100	30	235	0.2	16.94	410	24.20
PLA	T-02	80	50	210	0.2	17.784	413.56	23.25
ABS	T-12	80	50	235	0.2	18.998	390.83	20.57
PLA	T-03	60	60	210	0.2	18.762	305.65	16.29
ABS	T-13	60	32	235	0.2	19.89	382.36	19.22
PLA	T-04	40	30	210	0.2	20.191	275.84	13.66
ABS	T-14	40	30	235	0.2	19.68	349.78	17.77



3D SCANNERS IN MECHANICAL ENGINEERING

- Inspection of mechanical parts
- Machine design
- Automotive designing
- Rapid manufacturing and Reverse engineering

USEFULNESS OF 3D SCANNERS IN INSPECTION

- **CAD Analysis**
- **Dimensional Inspection**
- **GD&T Analysis**
- **Automated Inspection**
- **Product Development –** Embedding the scanned data in digital manufacturing process like 3D printing, product verification, engineering analysis helps to save time money and resources

DEVICES USED IN PROPOSED WORK

PROPELLER

3D SCANNER

Camera resolution	1.3 mega pixel
Light source	White
Stand off distance	290-480 mm
Computer requirement	Windows 8,9,10, (8gb ram)
Power supply	40watt
Input voltage	12 volt
Load capacity	5 kg

×

CMM

Model no.	M574 Crsyta Plus
Working mode	Standard, Manual
Probe used	TP-8 -3553
Working type	Bridge
X axis range	500 mm
Y axis range	700 mm
Z axis range	400 mm
Resolution	0.0005 mm
Work table Material	Black granite
Work table size	764*1175 mm
Tapped insert	M8*1.25 mm
Max work piece height	590 mm
Work piece load	180 kg
Mass	646 kg
Dimension	143412052267 mm
Air supply pressure	0.35 MPa
Air consumption	50 L/min

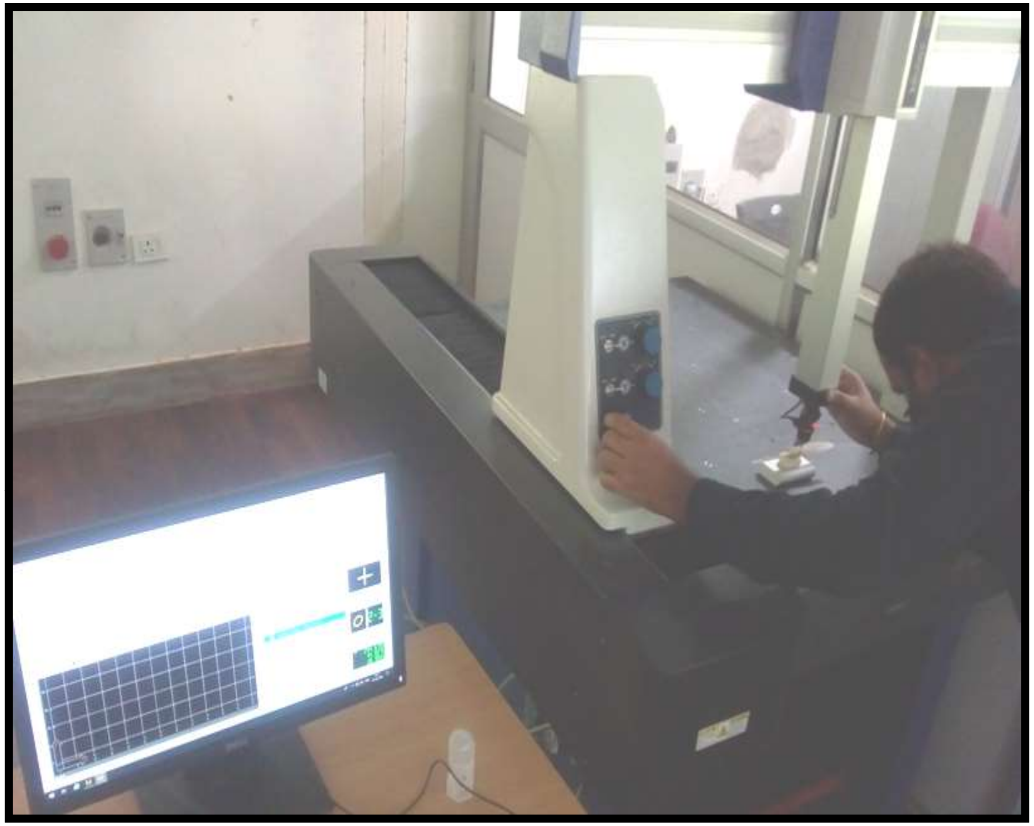
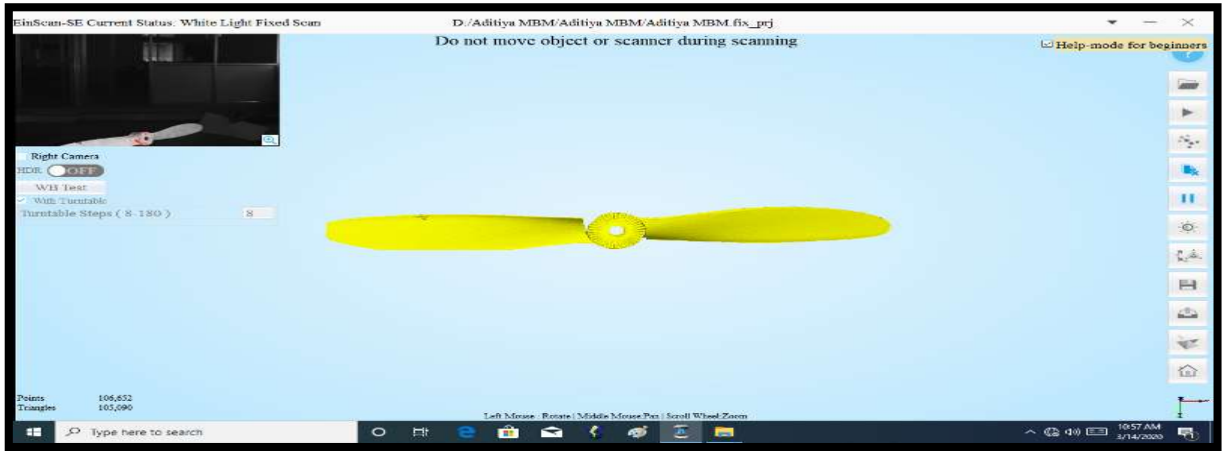


Fig: Coordinate Measuring Machine



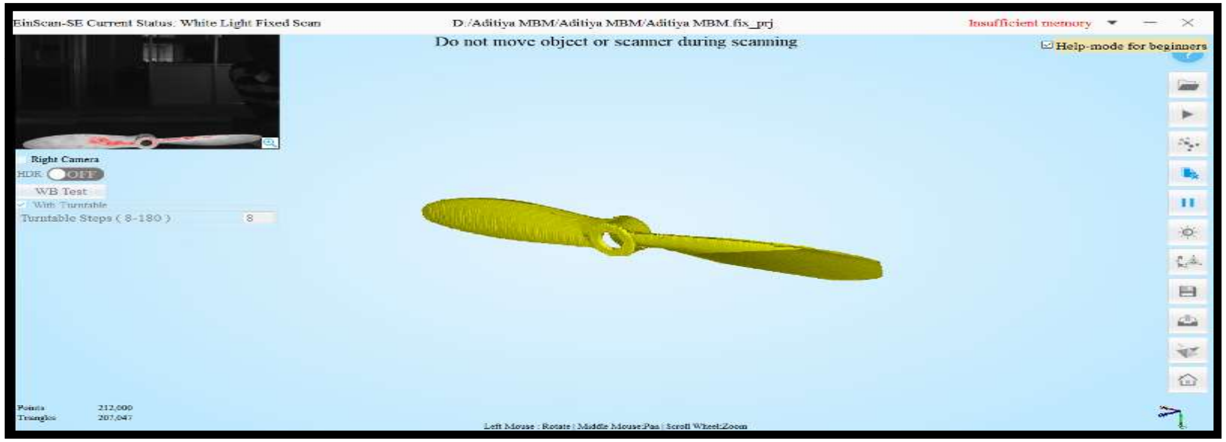


Fig: First and Second scan

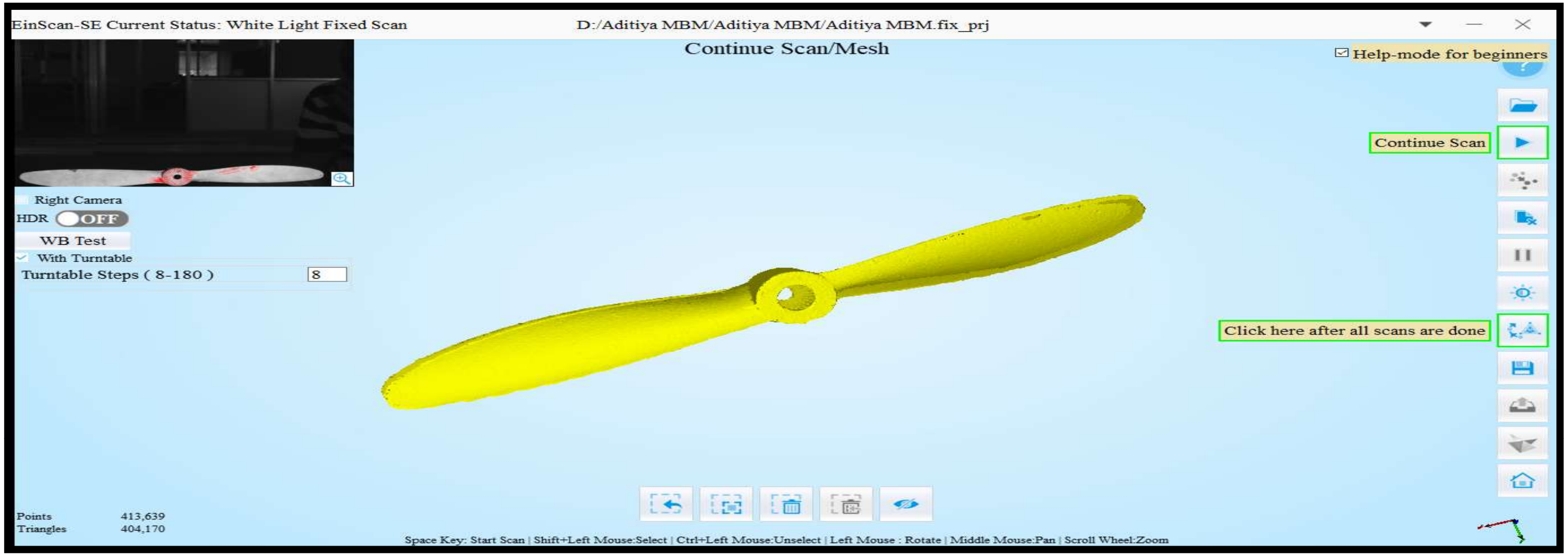


Fig: 3D Model obtained after processing of the point cloud data

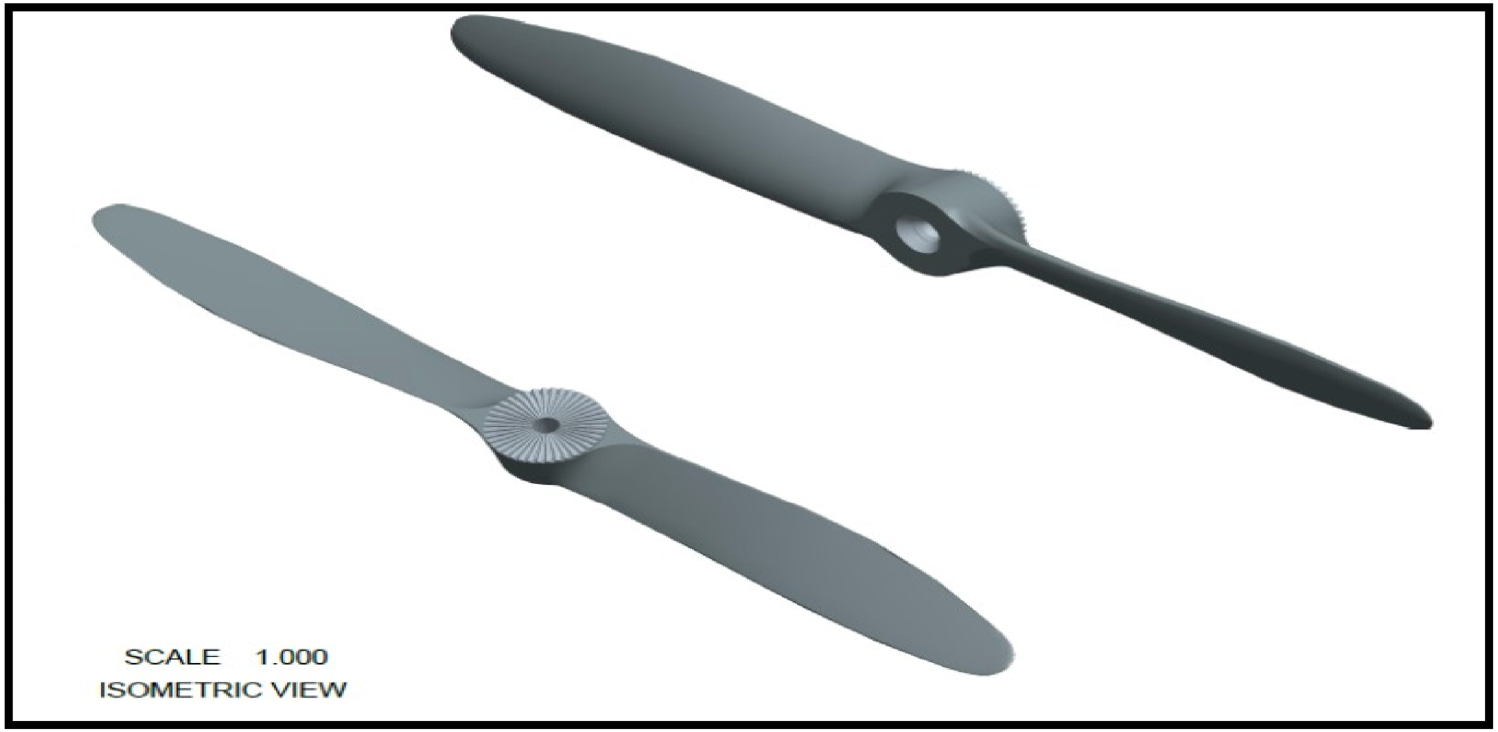


Fig: Final CAD model

Fig: Final 3D printed propeller

Fig: New modified propeller having three rotors

RESULTS

Results for Two Rotor Propeller

S.NO	PARAMETERS	INJECTION MOULDING	REVERSE ENGINEERING
1	Time elapsed to manufacture one propeller	1 hour	4 hour
2	Material employed	Nylon	PLA
3	Cost of material per unit gram	18 Rs	9 Rs
4	Cost of manufacturing one unit of propeller when production is less than 2.5 thousand units	180 RS	100 Rs
5	Cost of manufacturing one unit of propeller when production is more than 2.5 thousand units	80 Rs	100Rs

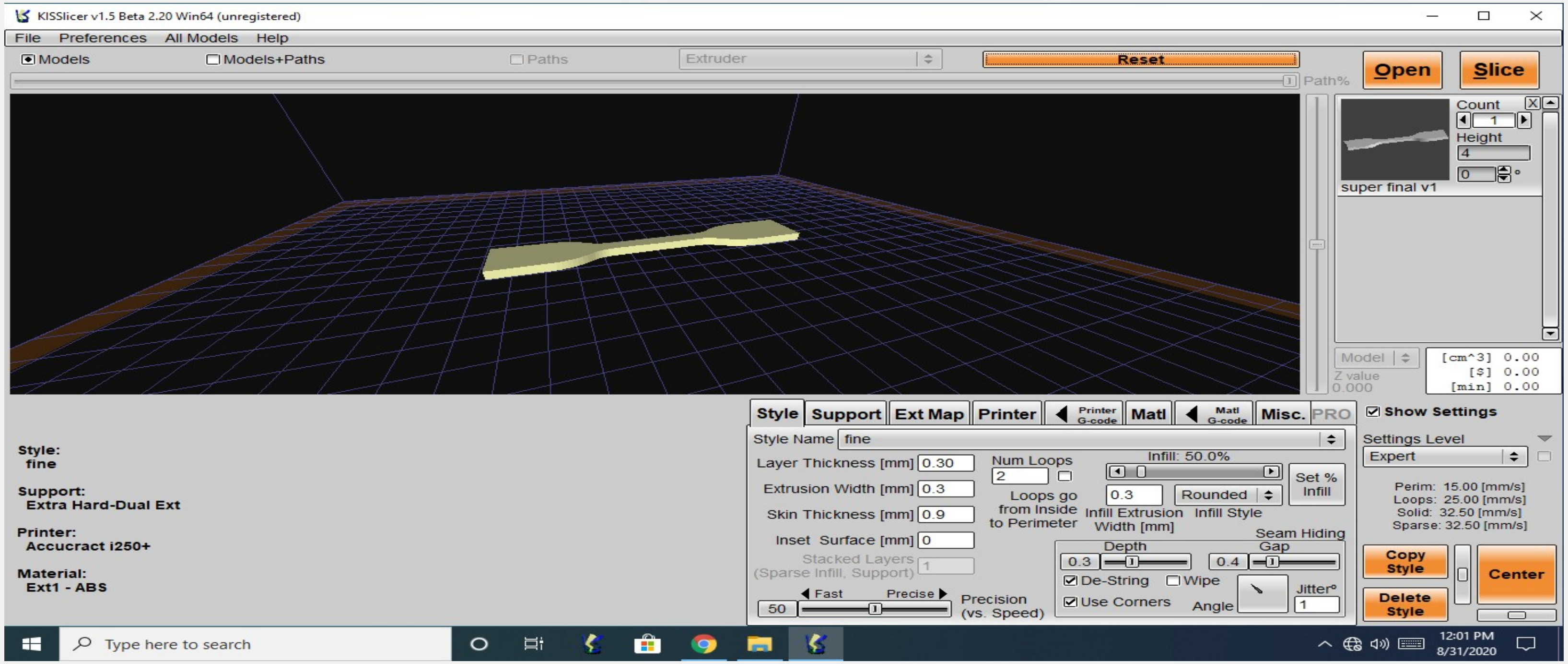
RESULTS OBTAINED FROM INSPECTION WITH 3D SCANNER

S.No	NOTATION	DIMENSION NAME	DIMENSIONS WITH 3D SCANNER (MM)	DIMENSIONS WITH CALLIPER (MM)	ACTUAL DIMENSIONS (MM)
1	L1	Length of propeller	202.5	203	203
2	L2	Length of each blade	89.12	89.35	90
3	D1	External diameter of hub	20.61	20	20
4	D2	Internal diameter of hub on back side	4.85	5	5
5	D3	Internal diameter of hub	10	10	10
6	H1	Total height of propeller hub	13.4	14.2	14
7	H2	Height of teethes on hub	1	1	1
8	T	Height of hub without teethes	11.78	12	12

Design of Experiment (DOE)

Experiment No.	Layer thickness (mm)	Infill style	Infill density (mm)
1	0.20	Straight	0.25
2	0.20	Octagonal	0.33
3	0.20	Rounded	0.50
4	0.25	Straight	0.33
5	0.25	Octagonal	0.50
6	0.25	rounded	0.25
7	0.30	Straight	0.50
8	0.30	Octagonal	0.25
9	0.30	Rounded	0.33

Axis	Most Influencing Factor	Contribution
X-axis	Infill Density	34.56%
Y-axis	Infill Style	63.62%
Z-axis	Layer Thickness	69.40%


KISSlicer v1.5 Beta 2.20 Win64 (unregistered)
File Preferences All Models Help
Models
Models+Paths
Paths
Extruder
Reset
Open
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Path%
Count
Height
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Style:
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Support:
Extra Hard-Dual Ext
Printer:
Accucract i250+
Material:
Ext1 - ABS
Style Support Ext Map Printer Printer G-code Matl Matl G-code Misc. PRO
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Extrusion Width [mm] 0.3
Skin Thickness [mm] 0.9
Inset Surface [mm] 0
Stacked Layers (Sparse Infill, Support)
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Loops go from Inside to Perimeter
Infill: 50.0%
Infill Extrusion Width [mm]
Rounded
Infill Style
Set % Infill
Seam Hiding
Depth
Gap
De-String
Wipe
Use Corners
Angle
Jitter°
Fast
Precise
Precision (vs. Speed)
Settings Level
Expert
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Solid: 32.50 [mm/s]
Sparse: 32.50 [mm/s]
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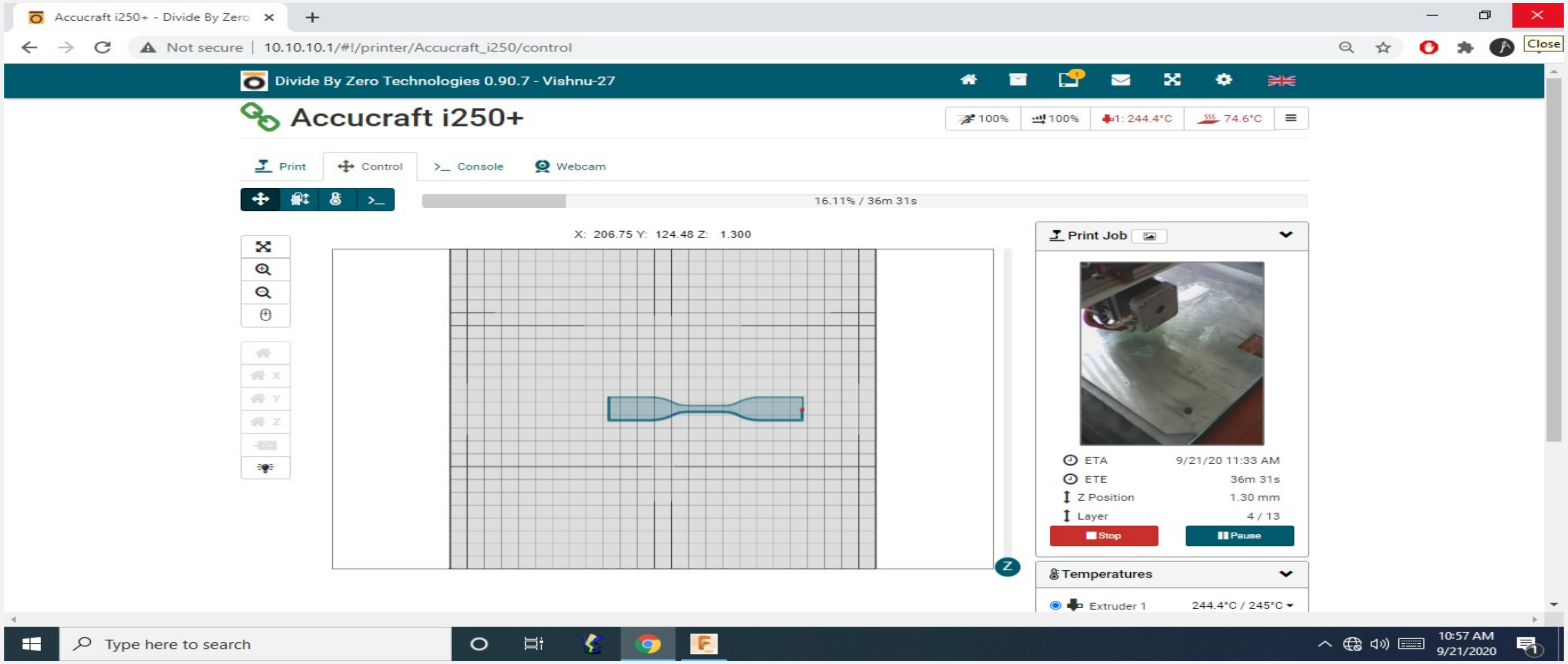
8/31/2020



Accucraft i250+ - Divide By Zero
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Not secure
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Print
Control
Console
Webcam
16.11% / 36m 31s
X: 206.75 Y: 124.48 Z: 1.300
Print Job
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Z Position 1.30 mm
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Stop
Pause
Temperatures
Extruder 1 244.4°C / 245°C
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9/21/2020


THANK YOU